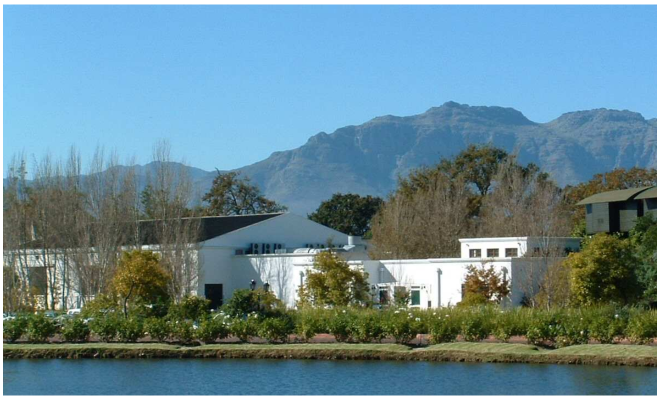
8th International Workshop on Finite Elements for Microwave Engineering

25-26 May 2006, Spier Wine Estate, Stellenbosch, South Africa



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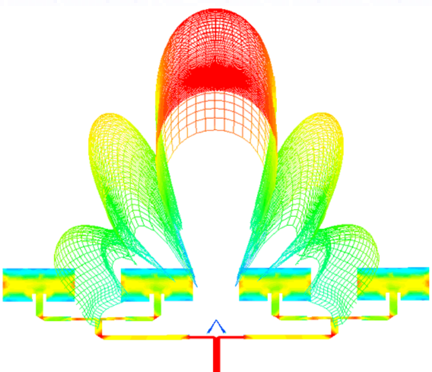

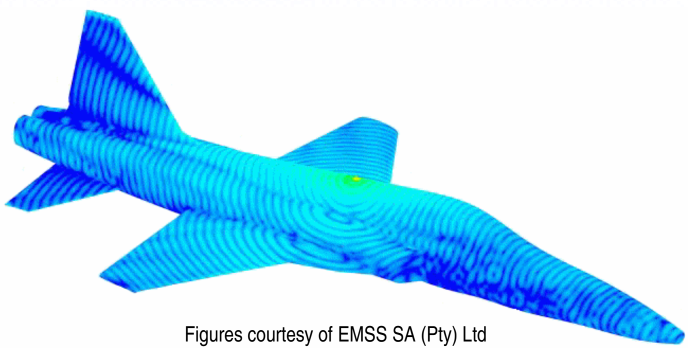

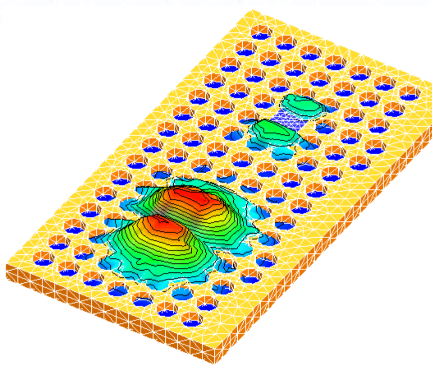

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General Chair’s Welcome

Dear Colleagues and Friends

On behalf of the organizing committee, it is a great pleasure to welcome you to my hometown of Stellenbosch, South Africa, for this the 8th International Workshop on Finite Elements for Microwave Engineering. This is the eighth in the series, dating back to 1992. The first workshop was entitled *Il metodo degli elementi finiti nelle applicazioni dell'elettromagnetismo* (National Workshop on Finite Elements in Electromagnetic Wave Problems) and was held in San Miniato (Pisa, Italy), May 26-27 1992. It was organized by Giuseppe Pelosi, who has continued to serve as General Co-Chair for all subsequent workshops. The 2nd workshop, held in Siena (Italy, 24-26 May 1994), was entitled the “International Workshop on Finite Elements in Electromagnetic Wave Problems”.

This highly-focussed biennial workshop has provided an ideal meeting place for researchers and practitioners active in the theory and application of the finite element method in RF and microwave engineering. Since its early inception in Italy, it has acquired a tradition of moving around the world; the 3rd workshop was scheduled for Halifax (Nova Scotia, Canada), 9-11 July 1996 - but was cancelled at the last minute (due to the deteriorating health of Peter Silvester); the 4th was held in Poitiers (France, 10-11July, 1998 – and from this workshop on, the name settled on the current “International Workshop on Finite Elements for Microwave Engineering”), the 5th in Boston (Massachussets, USA, 8-9 June 2000), the 6th in Chios (Greece, 30 May – 1 June 2002) and 7th in Madrid (Spain, 20-21 May 2004). This is the first time that the workshop is being held in the southern hemisphere, at the Spier Wine Estate outside Stellenbosch. Stellenbosch itself is the second oldest town in South Africa (dating back to the 1680's), the centre of the Western Cape winelands, and home of the University of Stellenbosch, the main organizer of the workshop.

The technical program features more than fifty papers, contributed from around the world, and provides an excellent overview of the current state-of-the-art in finite element research and application at RF and microwave frequencies. This field continues to grow in importance across a broad spectrum of electronic engineering applications. Many thanks to the session organizers who were instrumental in assembling the fine technical program. They are credited in this book of abstracts at each session. Thanks also to the members of the Steering Committee. We also welcome colleagues from applied mathematics at this workshop, and look forward to the additional theoretical insights they can offer.

From myself as General Chair, a special word of thanks to the Technical Program Chairman, Matthys Botha, and the Workshop Administrator, Hannelie van Wyk. These formal titles do not fully reflect the diverse fields in which they contributed to ensure the success of this event.

Ten years ago, Peter Silvester, one of the pioneers in this field, very sadly passed away at the peak of his professional life. At this workshop, we have a special session to commemorate his work, organized by his friend and colleague Ron Ferrari. Additionally, Ron has collated Peter's publications, which we include in this book of abstracts.

A new feature of this workshop is a plenary talk, which Raj Mittra has kindly agreed to present.

Many thanks also to our sponsors. Ansoft Corporation, EM Software & Systems, the European Office of Aerospace Research and Development (of the Air Force Office of Scientific Research, United States Air Force Research Laboratory), and the South African Joint AP/MTT chapter of the IEEE provided generous financial sponsorships. The IEEE Antennas and Propagation and Microwave Theory and Techniques societies, as well as the University of Florence, Italy, assisted with technical co-sponsorship. Rick Ziolkowksi, immediate Past President, is representing the IEEE APS at the workshop.

Many of you may be visiting South Africa for the first time - the local committee extends an especially warm welcome to you. Since the momentous political changes of the early 1990s, South Africa has again stepped forward onto all aspects of the world stage. Our country has much to offer visitors, and we hope you will enjoy the social events we have organized. These represent only a very small sample of the many attractions which have made South Africa in general and the Western Cape in particular one of the fastest-growing tourist destinations in the world over the last decade. Some of you are travelling further afield in South Africa following the workshop; we hope that your further travels will be memorable.

We trust at the end of the workshop it will be "*Totsiens*" (Afrikaans for "until we meet again") rather than goodbye, and look forward to welcoming you back in future years!

David B. Davidson
General Chair
May 2006

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Technical Program

Day 1 — Morning (Thursday, May 25)

Time	Events	
08:30	Registration	
09:00	Opening	
09:30	Session 1 — **Plenary** Organizers: *Davidson, D. B.; Pelosi, G.*	
09:30	Some Challenges in Computational Electromagnetics for Solving Large Problems and how we might meet Them *Mittra, R.*	
10:00	Session 2 — **FEM Applications: Antennas and Materials. Part 1** Organizers: *Kempel, L.; Meyer, F. J. C.*	Session 3 — **Adaptive Methods** Organizers: *Dyczij-Edlinger, R.; Webb, J. P.*
10:00	Periodic Array Modeling by the Hybrid Finite Element – Boundary Integral Technique Accelerated by a Multilevel Fast Spectral Domain Algorithm *Eibert, T. F.; Weitsch, Y.*	Adaptive Analysis with a Stationary FE-BI Formulation *Botha, M. M.; Jin, J.-M.*
10:20	Using the Efficiency of the Finite Element / Method of Moments Techniques for Establishing Safe Public and Occupation Zones around Base Station Antennas *Meyer, F. J. C.; Kellerman, V.*	On *hp*-Adaptivity for Maxwell's Equations: Error Estimation and Refinement Strategies *Ingelström, P.; Hill, V.; Dyczij-Edlinger, R.*
10:40	Modeling Electromagnetic Problems Using Hybrid FEM/MoM Technique *Sun, R.; Reddy, C. J.*	Adaptive Solution of Large Problems using Nearly Orthogonal Nedelec Bases in Multilevel Preconditioned Solvers *Peng, G.; Sun, D.-K.; Cendes, Z. J.*
11:00	Coffee Break	
11:40	Session 4 — **Multigrid- and Domain Decomposition Methods. Part 1** Organizers: *Cangellaris, A. C.; Lee, J.-F.*	Session 5 — **Advances in Absorbing- and Periodic Boundary Conditions** Organizers: *Botha, M. M.; Jin, J.-M.*
11:40	An Algebraic Multigrid Poisson Solver for On-Chip Interconnect and Cell Capacitance Extraction *Sumant, P.; Cangellaris, A. C.*	Implementation of Higher Order Absorbing Boundary Conditions in Vector Finite Element Methods *Gündeş, F.; Zhao, K.; Lee, J.-F.*
12:00	Domain Decomposition for Solving Large Electromagnetic Field Problems *Sun, D.-K.; Longtin, M. C.; Cendes, Z. J.*	Time-Domain Finite Element Simulation of Periodic Structures *Petersson, L. E. R.; Jin, J.-M.*
12:20	A Fast Domain Decomposition Method for Modeling Large Electromagnetic Problems *Zhao, K.; Lee, J.-F.*	An Extension of the Adaptive Absorbing Boundary Condition for the Tridimensional Maxwell Equations *Zerbib, N.; Bendali, A.*
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Day 1 — Afternoon (Thursday, May 25)

Time	Events	
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13:50	A novel class of differential form based interpolation operators for FEM *Bluck, M. J.; Walker, S. P.*	Eigen value Estimation of Finite Element-Boundary Integral Matrix Systems and their Application to FSS Design *Fischer, B. E.; Volakis, J. L.;Yagle, A. E.*
14:10	Decomposition of the mixed first-order, divergence-conforming function space on a tetrahedral mesh *Botha, M. M.; Davidson, D. B.*	Inhomogeneous Material Loading for Microstrip Leaky-wave Antennas *Jiang, H.; Pasala, K.; Kempel, L.*
14:30	Polygonal Finite Elements *Euler, T.; Schuhmann, R.; Weiland, T.*	Development of Time-Domain Finite Element Method for Analysis of Broadband Antennas and Arrays *Lou, Z.; Jin, J.-M.*
14:50	Curvilinear Vector Finite Elements using Hierarchical Basis Functions *Swartz, J. P.; Davidson, D. B.*	Finite Element Method for Periodic Structures *Yarga, S.; Sertel, K.; Volakis, J. L.*
15:10	A Singular Tetrahedral Finite Element for Sharp Metal Edges *Webb, J. P.*	
15:30	Coffee Break	
16:00	Session 8 — **Special Session in Honour of P. P. Silvester** Organizers: *Ferrari, R. L.*	
16:00	Papers by P.P.Silvester applying integral equation methods – a historical review *Ferrari, R. L.*	
16:20	The contribution of P.P. Silvester to Wave Electromagnetics *Pelosi, G.; Selleri, S.*	
16:40	The early contributions of P. P. Silvester to the finite element solution of saturable magnetic field problems *Webb, J. P.*	

Day 2 — Morning (Friday, May 26)

Time	Events	
09:00	Session 9 — **Time domain FEM. Part 1** Organizers: *Hesthaven, J. S., Rylander, T.*	Session 10 — **FEM Applications: Waveguides, Components, Active Devices, Lumped Elements** Organizers: *Mittra, R.; Selleri, S.*
09:00	Applying the Cement Technique to Non-Conformal Finite Element Time Domain Method for Electromagnetic Problems in R3 *Venkatarayalu, N.; Gan, Y. B.; Lee, J.-F.*	Simulating the Performance of Electronic Packages and Printed Circuit Boards *Bracken, J. E.; Polstyanko, S.; Cendes, Z. J.*
09:20	A Higher Order Discontinuous Galerkin Time Domain Method for Beam Dynamics Simulations in Linear Accelerators *Gjonaj, E.; Weiland, T.*	Implementation of Lumped Elements and Fast Frequency Sweep into the Vector Finite Element code for RF Electromagnetic Field Simulation *Grigoriev, A. D.; Salimov, R. V.; Tikhonov, R. I.*
09:40	Comparison of Time Domain FEM Formulations *Marais, N.; Davidson, D. B.*	A New FEM/BEM Domain Segmentation Approach For Solving High Diversity Electromagnetic Problems *Guarnieri, G.; Pelosi, G.; Rossi, L.; Selleri, S.*
10:00	Use of Hanging Variables in the Time Domain for Adaptive Refinement and Subgridding *Srisukh, Y.; Lee, R.; Venkatarayalu, N.; Gan, Y. B.*	Modeling Wave Guiding Structures at Very Low Frequencies *Lee, S.-C.; Lee, J.-F.*
10:20	Discontinuous Galerkin Time Domain FEM for Solving Maxwell Equations *Wong, M.-F.; Wiart, J.*	Model Order Reduction of Finite Element Models of Electromagnetic Structures with Embedded Frequency-Dependent Multiports *Wu, H.; Cangellaris, A. C.*
10:40	Coffee Break	
11:20	Session 11 — **Time domain FEM. Part 2** Organizers: *Hesthaven, J. S., Rylander, T.*	Session 12 — **Hybrid Methods** Organizers: *Lee, R.; Volakis, J. L.*
11:20	Simulating the Behavior of Metamaterials and their Applications: Time Domain Advantages and Disadvantages *Ziolkowski, R. W.*	A Non Standard Fast Multipole Finite Element Method for Scattering and Radiation Problems *Barrio-Garrido, R. M.; García-Castillo, L. E.; Gómez-Revuelto, I.; Salazar-Palma, M.*
11:40	A Novel Time Domain Approach for Modeling the Coupling Problem between Two Large Arrays Mounted on a Complex Platform and Separated by a Large Distance *Mittra, R.; Abd-El-Raouf, H. E.; Huang, N.-T.*	On the Formulation of the 3D Hybrid Finite Element - Boundary Integral Technique: … Still the Old One: Formulation, Solution, Results *Eibert, T. F.; Tzoulis, A.*
12:00	Stable interfaces between the time-domain FEM and the FDTD scheme *Degerfeldt, D.; Rylander, T.*	Hybrid MoM-FDTD for Human RF Exposure Analysis *Lacroux, F.; Wong, M.-F.; Wiart, J.*
12:20	Yee's Scheme for the Solution of Maxwell's Equation on Unstructured Meshes *Hassan, O., Sazonov, I.; Morgan, K.; Weatherill, N. P.*	Modeling Large Phased Array Antennas Using the Finite Difference Time Domain Method and the Characteristic Basis Function Approach *Mittra, R.; Farahat, N.; Huang, N.-T.*
12:40	Lunch Break	

Day 2 — Afternoon (Friday, May 26)

Time	Events	
13:50	Session 13 — **Advances in FEM Formulations and Solvers** Organizers: *Botha, M. M.; Jin, J.-M.*	Session 14 — **Optimization Techniques, Parameter Space Sweep. Part 1** Organizers: *Pelosi, G.; Zapata, J.*
13:50	Homogenization of Large Screens in Finite Element Analysis *Bardi, I.; Cendes, Z. J.*	Order Reduction of Finite Element Models Parameterized by Multivariate Polynomials *Farle, O.; Hill, V.; Ingelström, P.; Dyczij-Edlinger, R.*
14:10	SPICE-Compatible Finite Element Models of Maxwell's Equations *Cangellaris, A. C.*	A Reduced Order Model for the Rapid Computation of Monostatic Scattering Width Distributions *Ledger, P. D.; Morgan, K.*
14:30	An Efficient Solver for High-Order Vector Finite Elements *Ingelström, P.; Hill, V.; Dyczij-Edlinger, R.*	Genetic Optimization of Unconventional Inductive Frequency Selective Surfaces by Using a Hybrid Mode Matching - FEM technique *Pellegrini, A.; Monorchio, A.; Manara, G.*
14:50	Coffee Break	
15:20	Session 15 — **Multigrid- and Domain Decomposition Methods. Part 2** Organizers: *Cangellaris, A. C.; Lee, J.-F.*	Session 16 — **Optimization Techniques, Parameter Space Sweep. Part 2** Organizers: *Pelosi, G.; Zapata, J.*
15:20	Adaptive Domain Decomposition Techniques in Electromagnetic Field Computation *Hoppe, R. H. W.*	FEM Optimization of Coplanar-to-Coplanar waveguide transitions in Flip-Chip Interconnections *Agastra, E.; Limiti, E.; Pelosi, G.; Selleri, S.*
15:40	A Nonuniform hp-Finite Element Method for the Computation of Electromagnetic Cavity Modes *Nickel, P.; Hill, V.; Ingelström, P.; Dyczij-Edlinger, R.*	Optimization of Devices Based on Bodies of Revolution with Finite Elements *Martínez-Fernández, J.; Monge, J.; Zapata, J.; Gil, J. M.*
16:00	Studies of Numerical Error Introduced by the Cement Variables in the Domain Decomposition Method *Rawat, V.; Lee, J.-F.*	Parametric FEM Modal Analysis in microwaves and EMC *Wong, M.-F.; Gati, A.; Orjubin, G.; Richalot, E.; Picon, O.; Wiart, J.; Hanna, V. F.*

Session 1 — Plenary

Organizers

Davidson, D. B.; Pelosi, G.

Presentations

1. Some Challenges in Computational Electromagnetics for Solving Large Problems and how we might meet Them
   Mittra, R.

SOME CHALLENGES IN COMPUTATIONAL ELECTROMAGNETICS FOR SOLVING LARGE PROBLEMS AND HOW WE MIGHT MEET THEM

Raj Mittra
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Spectacular increase in our capability to model, simulate the performance, and design complex electromagnetic systems on the computer in recent years has fueled our desire to solve even larger and more complex problems than we have been able to in the past, so that we can say with confidence that the electromagnetic system we develop for a given application will be "correct by design." Of course, there are many competing CEM approaches at our disposal, and we have recently witnessed great progress in enhancing these numerical techniques in several different ways. These include, for example, employing the Fast Multipole Method (FMM) to speed up the Method of Moments (MoM); hybridizing the Finite Element Method (FEM) as well as the asymptotic methods with the MoM; and, using the conformal Finite Difference Time Domain (FDTD) method to handle arbitrarily shaped objects without staircasing. There are several excellent commercial CEM tools available in the market today that are routinely being used for modeling and simulation of a wide variety of electromagnetic structures, and in recent years several of them have successfully incorporated these enhancements to enlarge their problem-solving capabilities and/or reducing the CPU time. Their unique advantages as well as limitations are well known and a discussion of these features is beyond the scope of this presentation. Instead, we will focus here on identifying a whole host of "Challenging Problems" in computational EM, all of which belong to the category of real-world problems of practical interest. Another common feature, shared by the problems belonging to this group, is that they are characterized by a very large number of DOFs (degrees of freedom), 10^9 or even greater for instance, typically beyond the range of many, if not most available CEM solvers. And yet, they are sufficiently complex and inhomogeneous in nature, often because they contain metamaterials and articulated RAMs that renders them unsuitable for analysis via the asymptotic techniques such as the ray methods, and they are only amenable to attack via numerically rigorous techniques.

Some examples of such complex problems include large phased arrays comprising of thousands of elements and/or mounted on curved surfaces, with or without FSS radomes that may also be curved; coupling between large antennas operating in topside environment; wearable antennas and prediction of performance characteristics as well as the biohazard aspects of these antennas; modeling of through-wall imaging systems; and multilayer, electronic packaging and board design to name a few.

The presentation would then discuss a trend in CEM that is significantly impacting the computing landscape, namely the use of highly parallel computers, and end with the prediction that only those solver that can adapt to this trend would be the winners, if not the survivors, in the large-scale computing enterprise.

Session 2 — FEM Applications: Antennas and Materials. Part 1

Organizers

Kempel, L.; Meyer, F. J. C.

Presentations

1. Periodic Array Modeling by the Hybrid Finite Element – Boundary Integral Technique Accelerated by a Multilevel Fast Spectral Domain Algorithm
   Eibert, T. F.; Weitsch, Y.

2. Using the Efficiency of the Finite Element / Method of Moments Techniques for Establishing Safe Public and Occupation Zones around Base Station Antennas
   Meyer, F. J. C.; Kellerman, V.

3. Modeling Electromagnetic Problems Using Hybrid FEM/MoM Technique
   Sun, R.; Reddy, C. J.

Periodic Array Modeling by the Hybrid Finite Element – Boundary Integral Technique Accelerated by a Multilevel Fast Spectral Domain Algorithm

Thomas F. Eibert and Yvonne Weitsch
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The infinite periodic array model is still an important approach for analysis and design of large finite arrays. With periodic excitation, the field solution in the array becomes also periodic and Floquet's theorem can be employed to reduce the computational domain down to a single array element (unit cell), thus significantly speeding up analysis and reducing memory requirements. Arbitrarily inhomogeneos array elements can be modelled by the hybrid finite element - boundary integral (FEBI) technique. The spectral domain (SD) Floquet mode expansion leads to diagonalization of the BI and this property can be utilized to efficiently evaluate the matrix-vector product contributions pertinent to the BI required within iterative equation solvers. Similar to the Multilevel Fast Multipole Method (MLFMM), a Multilevel Fast Spectral Domain Algorithm (MLFSDA) is constructed and employed to accelerate a FEBI technique working with tetrahedral volume and triangular BI elements as well as a stand-alone integral equation technique for the modelling of planar metallic array elements within a multilayered medium.
With MLFSDA, unit cells with dimensions of many wavelengths can be efficiently handled. Thus, it is possible to model multilayer frequency selective surfaces with dissimilar periodicities by the supercell approach, where a period common to all individual layers is defined and several array elements in the various layers are placed in the computational unit cell.
With the capability to handle large unit cells, it becomes possible to place entire finite configurations in a periodic unit cell, which is large enough to decouple the periodic images. Thus, the advantages of the periodic modelling approach (Floquet mode series instead of SD integral) can also be used to compute finite configurations such as finite antenna arrays. Various computational results involving frequency selective surfaces and antenna applications are shown.

Using the Efficiency of the Finite Element / Method of Moments Techniques for Establishing Safe Public and Occupation Zones around Base Station Antennas

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Abstract:

Specific Absorption Rate (SAR) calculations or measurements can be used for accurate human exposure assessment in close proximity to base station antennas [1]. SAR results can then be compared to international guidelines and standards for safe human exposure to radiofrequency fields [2]. From this the **public** and **occupational** compliance zones around base stations can be obtained. Extensive work is required to accurately measure or calculate the SAR from commercial base station antennas. In the case of measurements, expensive equipment and an established measurement setup is required. For computations, comprehensive validation of the antenna models is required and complex full wave techniques must be employed for the analyses. Both these routes are too elaborate if SAR is to be used as the basis for compliance zone calculations in a practical mobile phone network. In this paper we show that very simple analytical formulas can be used for compliance zone calculations around GSM900, GSM1800/1900 and UMTS base station antennas. This concept was first proposed by Nortström in [3] for UMTS. We have expanded on this idea to take into account different frequencies, phantom shape / size, phantom orientation, multiple sources and off-broadside positions. This all required extensive simulations and the hybrid FEM / MoM technique proved to be very efficient for these "bulk" simulations.

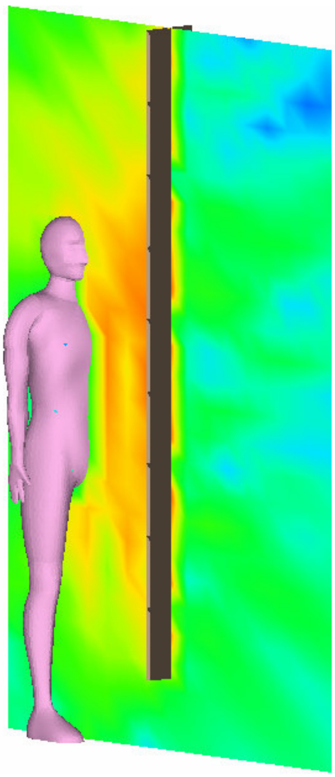

Figure 1: Field distribution in and around a human body calculated using the FEM/MoM technique

References:

[1] FJC Meyer, MJ van Wyk and RA Kellerman, "*Numerical Compliance Zone Profiling of a Commercial Base Station Antenna,"* in Proceedings of the Twenty-Seventh Annual Meeting of the Bioelectromagnetic Society, (Dublin, Ireland), June 2005.

[2] "*ICNIRP Guidelines for limiting exposure to time-varying electric, magnetic, and electromagnetic fields (up to 300GHz)*," Health Physics, vol. 74, no. 4, pp. 494-522, April 1998.

[3] E Nordström, "*Calculation of specific absorption rate (SAR) for typical WCDMA base station antennas at 2140 MHz*", Master of Science thesis at Department of Physics and Measurement Technology, Biology and Chemistry, Linköping Institute of Technology, performed at Ericsson AB Ericsson Research, EMF Health and Safety, December 2004.

Modeling Electromagnetic Problems using Hybrid FEM/MoM Technique

Rensheng Sun and C. J. Reddy

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Abstract:

The finite element method (FEM) is widely used in computational electromagnetics since it is very efficient on modeling 3D volumes that have complex geometrical details and material inhomogeneities. For radiation/scattering problems, usually the radiation condition in FEM implementations is applied using appropriate absorbing boundary condition (ABC), or perfectly matched layer (PML) technique. This requires the domain of discretization be extended far from the source region. The integral equation method such as Method of Moments (MoM) is an ideal formulation for open boundary radiating/scattering structures. As a result, the domain of discretization can be kept to a minimum. The hybrid FEM/MoM has recently been introduced with the commercial code FEKO suite 5.0. This development exploits the benefits of both techniques and enables FEKO to solve certain classes of electromagnetic problems with optimal efficiency. As an example, it was employed for specific absorption rate (SAR) calculation in a human phantom in the near field of a typical base-station antenna [F.J.C. Meyer, *etc*. "Human exposure assessment in the near field of GSM base-station antennas using a hybrid finite element/method of moments technique," IEEE Trans. Biomedical Engineering, vol. 50, no. 2, pp. 224-233, Feb. 2003].

In this paper, we address application of hybrid FEM/MoM technique in FEKO for a class of radiation (antenna structures) and scattering problems. As a first step, hybrid FEM/MoM technique in FEKO has been validated using a slotted waveguide problem (Fig. 1). We also exploited various hybrid techniques in FEKO (Physical Optics, Uniform Theory of Diffraction) to incorporate electrically large structures in conjunction with application of hybrid FEM/MoM technique for antennas mounted on automobiles, ships etc. Various examples and numerical results will be presented at the workshop.

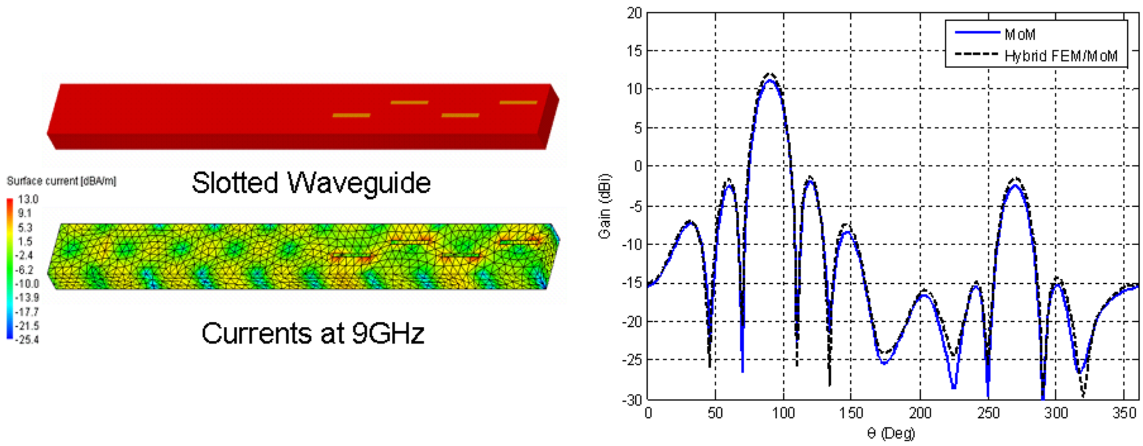

Fig.1. Analysis of slotted waveguide at 9GHz.

Session 3 — Adaptive Methods

Organizers

Dyczij-Edlinger, R.; Webb, J. P.

Presentations

1. Adaptive Analysis with a Stationary FE-BI Formulation
   Botha, M. M.; Jin, J.-M.

2. On hp-Adaptivity for Maxwell's Equations: Error Estimation and Refinement Strategies
   Ingelström, P.; Hill, V.; Dyczij-Edlinger, R.

3. Adaptive Solution of Large Problems using Nearly Orthogonal Nedelec Bases in Multilevel Preconditioned Solvers
   Peng, G.; Sun, D.-K.; Cendes, Z. J.

Adaptive analysis with a stationary FE-BI formulation

Matthys M. Botha[*] and Jian-Ming Jin[†]

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A comprehensive adaptive finite element-boundary integral (FE-BI) analysis scheme for the time-harmonic, electromagnetic analysis of three-dimensional inhomogeneous scatterers/radiators in free-space, is presented [1]. The adaptive scheme is based upon a recently proposed symmetric FE-BI formulation (with underlying variational principle) which yields electric and magnetic field solutions simultaneously [2].

The adaptive scheme employs a posteriori error estimates which exploit the availability of both field solutions, and estimates error distributions and global solution quality for the electric and magnetic fields separately. The scheme automatically determines which elements should be refined in order to equi-distribute the estimated error, given the requested type of refinement (h, p or hp). This automatic determination is based on extrapolating the elemental error estimates, using interpolation error estimates. The algorithm terminates when specified tolerance levels are reached by the electric and/or magnetic field global solution quality estimates (these may be specified independently). The only required user specifications within the algorithm are the termination tolerances and the types of refinements to effect.

Results are presented to demonstrate the significant extent to which computational costs may be reduced through use of the adaptive scheme, within the scope of the presented error measures. The proposed scheme could also be used together with other types of error estimates and it could be adjusted to other FE or FE-BI formulations.

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On hp-Adaptivity for Maxwell's Equations: Error Estimation and Refinement Strategies

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We report on ongoing work in the development of an hp-adaptive finite element solver for Maxwell's equations. In particular we focus on error estimation and strategies for refinement and unrefinement.

A posteriori error estimates are used to determine which elements to refine in standard h- or p-adaptive schemes. In hp-adaptive schemes, we need not only determine which elements to refine, but also how to refine these elements. To tackle this problem, we present two different error estimates, which estimates the effects of h-refinement and p-refinement, respectively. Further, to be able to decide how to refine in order to maximize the convergence rate, we also give an estimate for how various refinements affect the computational cost.

The error estimates we present are goal-oriented, that is, they estimate the error of a desired output, for example scattering parameters. They are based on the Dual-Weighted Residual (DWR) method [1], which require an approximate solution to a dual problem in an augmented finite element space. An estimate for the effect of p-refinement is obtained when the finite element space is enriched by higher order hierarchical basis functions. Similarly, an estimate for the effect of h-refinement is obtained by using an h-refined grid for the dual solution.

The main part of this presentation concerns how to efficiently compute approximate dual solutions by solving local problems, and how to set up and solve these local problems efficiently. In particular, we show how to take great advantage of the similarity between the refined elements and their parents when setting up the local problems for the h-refined mesh.

Algorithms and analysis are based on tetrahedral meshes and the hierarchical incomplete order $\mathbf{H}$(curl)-conforming basis in [2]. This basis spans the same spaces as those defined by Nédélec in [3]. As a consequence, it is possible to exactly represent a coarse grid solution on a fine grid using basis functions of the same order – and this is done using precomputed local inter-grid operators [4]. We note that this is in general not possible for some other incomplete order bases that have been presented in the literature.

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Adaptive Solution of Large Problems using Nearly Orthogonal Nedelec Bases in Multilevel Preconditioned Solvers

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A set of nearly orthogonal Nedelec bases for use with multilevel preconditioned solvers was presented in [1]. This procedure uses different orders of $H^P(curl)$ hierarchical basis functions to generate an efficient high-order solution algorithm on a single grid. It casts the system matrix in block form by employing a set of hierarchical vector basis functions with maximum orthogonality. The preconditioner for the first multi-level is obtained by multifrontal LU decomposition of $H^0(curl)$ with thresholding. Writing the system matrix in a two-level recursive form, Schur decomposition is used to compute higher-level solutions. These higher-level solutions are obtained using the conjugate gradient method with multiplicative preconditioning.

A key advantage of the multilevel preconditioned solver is a linear growth rate in memory requirements and a near linear growth rate in solution time. Figure 1 shows the performance of the new multilevel preconditioned solver compared to a traditional direct multifrontal solver. This near linear growth rate makes the algorithm ideal for solving large problems. Figure 2 shows two such problems involving antenna placement on an airplane and determining the radiated emissions from a 2.4 GHz cell phone in a car.

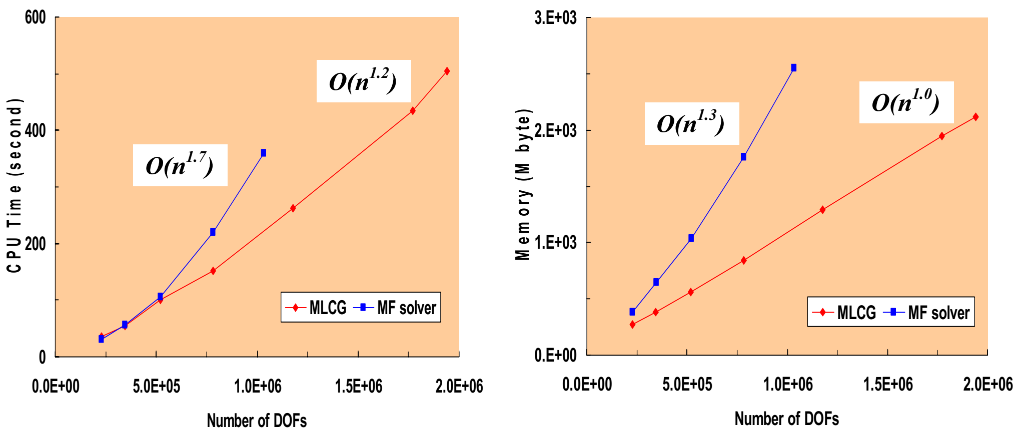

Figure 1. Memory and CPU time of the multilevel preconditioned solver vs. a traditional multifrontal solver.

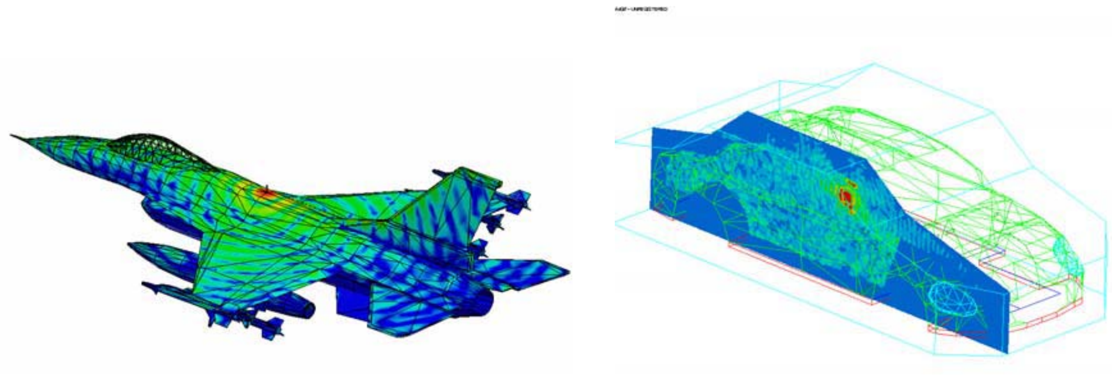

Figure 2. Radiation from an antenna on an airplane and from a cell phone in a car.

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Session 4 — Multigrid- and Domain Decomposition Methods. Part 1

Organizers

Cangellaris, A. C.; Lee, J.-F.

Presentations

1. An Algebraic Multigrid Poisson Solver for On-Chip Interconnect and Cell Capacitance Extraction
   Sumant, P.; Cangellaris, A. C.

2. Domain Decomposition for Solving Large Electromagnetic Field Problems
   Sun, D.-K.; Longtin, M. C.; Cendes, Z. J.

3. A Fast Domain Decomposition Method for Modeling Large Electromagnetic Problems
   Zhao, K.; Lee, J.-F.

An Algebraic Multigrid Poisson Solver for On-Chip Interconnect and Cell Capacitance Extraction

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Accurate extraction of on-chip interconnect and cell capacitance is an essential enabling capability for one of the most important classes of IC electronic design automation tools, namely, timing simulators. While a few technology generations back integral equation solvers, accelerated through the use of fast iterative solution processes such at the fast multipole method, were championed as the most efficient solvers for on-chip capacitance extraction, random walk-based solvers have recently become the solvers of choice for on-chip capacitance extraction. In addition to competitive computational efficiency, the two key additional features that make such solvers preferable to integral equation solvers are: a) the fact that the statistical nature of the underlying methodology used yields readily error bounds in the calculation of the net capacitance; b) their ability to handle more efficiently and with better accuracy than integral equation techniques, multiple conformal dielectrics, non-planar metallization and metal fills.

When it comes to the latter attribute, one could argue that finite element methods have an advantage. This is especially true for the latest and future technologies, where extensive use is made of multiple material layers in support of the reliable manufacturing of a few-tens-of-nanometers thick, metal features in low-dielectric-constant insulating media. While such material complexity is more easily handled by a finite element solver than a random walks-based one, the computational efficiency of the latter is anticipated to be superior despite the algorithmic overhead associated with the handling of integration paths that cross dielectric interfaces. The primary reason for this is the significant disparity in the spatial scales encountered in the on-chip metallization topography (low-submicron cross sectional dimensions of wires of length in the order of millimeters) and their impact on the number of degrees of freedom in the finite element model.

In this paper we will report on our experience with the application of algebraic multigrid techniques for expediting the iterative solution of a finite element Poisson solver, streamlined for on-chip capacitance extraction. The geometric and material attributes of the boundary value problem under investigation offer an ideal framework for assessing how algebraic multigrid methods fair when used in conjunction with finite element models of high material complexity.

Domain Decomposition for Solving Large Electromagnetic Field Problems

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Finite element domain decomposition methods for solving large electromagnetic field problems are presented. In domain decomposition, a large computational domain is divided into subdomains, a procedure employed to ensure field continuity across subdomain boundaries, and the finite element matrix in each subdomain is assembled and solved separately. Domain decomposition starts by splitting the solution domain into subdomains (Ω_1, Ω_2…). A mesh of tetrahedral elements is defined within each subdomain. The FEM equations are obtained from the vector wave equation using Galerkin's method

$$< \nabla \times \vec{E}_a^*, \frac{1}{\mu_{ra}} \nabla \times \vec{E}_a > - k^2 < \vec{E}_a^*, \varepsilon_{ra} \vec{E}_a > + \gamma_a < \vec{E}_a^*, \vec{E}_a >_\Gamma =$$
$$- jk\eta < \vec{E}_a^*, \vec{J}_a > + c_E < \vec{E}_a^*, \vec{E}_{tb} > + c_H < \vec{E}_a^*, \vec{H}_b \times \vec{n}_b >$$

The resulting matrix equation for each domain depends on the unknowns on the surfaces (S_1, S_2,…) and any excitations

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} + \gamma_a C_{aa} \end{bmatrix} \begin{pmatrix} e_a^I \\ e_a^\Gamma \end{pmatrix} = \begin{pmatrix} b_a^I \\ b_a^m \end{pmatrix}$$

Here, C_{aa}, the interface matrix, is the projection of the mesh elements from one face onto the corresponding mesh elements in the adjacent subdomain. For repeated subdomains, where the geometry in each subdomain is identical, this computation is performed only once and the matrix solutions are reused. The fields in the subdomains are glued together using Robin transmission conditions

$$\vec{H}_a \times \vec{n}_a + \gamma_a \vec{E}_{ta} = -\vec{H}_b \times \vec{n}_b + \gamma_a \vec{E}_{tb} \text{ on } \Gamma_{ab}$$
$$\vec{H}_b \times \vec{n}_b + \gamma_b \vec{E}_{tb} = -\vec{H}_a \times \vec{n}_a + \gamma_b \vec{E}_{ta} \text{ on } \Gamma_{ba}$$

The field quantities on the boundaries (S_1, S_2…) are calculated iteratively. Once these boundary fields are known they can be used to compute the fields within each domain. Therefore, in domain decomposition, the unknowns within each subdomain are computed iteratively, and the large matrix that would result from an entire domain solution is never created. After the iterations converge, the field solution for the entire model is known and the scattering matrix and/or antenna parameters can be computed.

As an example, consider the Rotman lens in Figure 1(a). Here four independent subdomains are used in the solution, some of which were repeated to provide a total of nine subdomains. Notice that the fields are continuous across subdomain boundaries. A traditional entire domain solution yielded near identical results. In fact, Figure 1(b) presents the visually identical results of computing the array factors by using domain decomposition and by using entire domain simulation. In this case, only 650MB was required to compute the domain decomposition solution versus almost 1.8GB required for the entire domain solution.

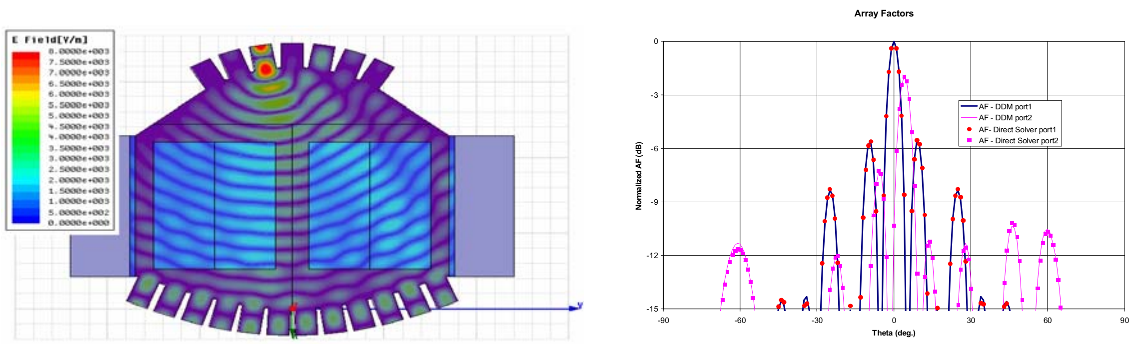

Figure 1(a). Radiation through a Rotman lens; (b) the lens array factor.

A Fast Domain Decomposition Method for Modeling Large Electromagnetic Problems

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Computational Electromagnetic (CEM) plays a crucial role in accurate predictions of design performance prior to construction. One of the important CEM applications includes problems with finite number of different building blocks. The examples of such problems contain antenna arrays, frequency selective surfaces, anechoic and semi-anechoic chambers, and metamaterials. This paper proposes a non-conforming and non-overlapping domain decomposition method (DDM) to take the advantage of periodicity in such structures, which at the same time is general enough to analyze arbitrary geometries. The flexibility of non-conforming triangulations across sub-domains not only relaxes the mesh generation and adaptive mesh refinement process, but more importantly avoids the periodic mesh constraints on the sides of the repeating blocks. A popular approach to address the non-matching grids issue in the non-overlapping DDM is the cement method, which accounts for the non-conformity between domains via a set of cement variables. Unlike the traditional mortar method, the current approach introduces an additional variable possessing physical meaning, and resulting in non-zero diagonal block in the final matrix equation.

Since DDM allows the independent handling of different domains, one finds that some of the domains can be easily changed without affecting the other domains. Consequently, any information or data for the unchanged domains can be retained and reused in any other instances. Extending this idea, it would be very beneficial to employ the finite element tearing and interconnecting (FETI) technique if a domain should be reused many times. FETI algorithm results in a “transfer function” or a numerical Green’s function as a matrix form which, once obtained, can be readily combined with other domains in the solution process of DDM as a matrix-vector multiplication form. Through FETI process, the information in the entire volume of a domain is translated into the information on the boundary surface. The construction of the numerical Green’s function is mainly accomplished through three major ingredients; 1) Decouple domains by the use of the FETI algorithm; 2) Solve the resulting finite element matrix equation with multiple right-hand-sides using state-of-art pMUS preconditioner; 3) Compress the numerical Green’s function by employing an efficient rank-revealing SVD algorithm.

Session 5 — Advances in Absorbing- and Periodic Boundary Conditions

Organizers

Botha, M. M.; Jin, J.-M.

Presentations

1. Implementation of Higher Order Absorbing Boundary Conditions in Vector Finite Element Methods
   Gündeş, F.; Zhao, K.; Lee, J.-F.

2. Time-Domain Finite Element Simulation of Periodic Structures
   Petersson, L. E. R.; Jin, J.-M.

3. An Extension of the Adaptive Absorbing Boundary Condition for the Tridimensional Maxwell Equations
   Zerbib, N.; Bendali, A.

Implementation of Higher Order Absorbing Boundary Conditions in Vector Finite Element Methods

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Many problems of interest including electromagnetic scattering and radiation problems require the solution of Maxwell's Equations in an infinite domain. However due to the limitations on the CPU time and computer memory, it is impossible to inherit the infinite nature of these problems directly into the numerical solution. Absorbing Boundary Conditions (ABC) is one of the methods used to overcome this difficulty when solving open region problems with the Finite Element Method (FEM).

Physically, the solution of an open region problem is expected to resemble the waves propagating to infinity. Whereas, in using FEM, it is compulsory to bound the computational domain with artificial truncation surfaces. ABC is the mathematical condition imposed on these surfaces to include the effect of infinite propagation into the solution. For obtaining accurate results, it is necessary to use appropriate mesh truncation schemes along with accurate ABCs which decrease the amount of field artificially reflected from the boundary surface. However, increasing the accuracy of the ABC gives rise to an increase in the computational complexity. Because of this reason, previously derived ABCs had difficulties in implementation which lead either to destruction of sparsity and symmetry of the FEM matrix or reduction of accuracy. The ABC presented here is capable of producing credible numerical results and also can easily be implemented via introduction of a cement variable on the surface it is imposed.

The derivation of the proposed ABC starts with the Maxwell's Curl Equations in source-free medium. Writing these equations in terms of the components of **E** and **H** in the Cartesian Coordinate System and assuming a z-directed wave, it is possible to construct a system of equations. Solving this system gives an expression which is to be called the transparent condition (TC). This condition is exact and it is satisfied for any wave incident upon a planar surface and directed along the normal of the surface. ABC is obtained by approximating the terms in the Taylor Expansion of TC. For a first order approximation, the ABC obtained provides an accuracy equivalent to that of second order Padé condition. It is possible to write a symmetric variational formulation with this second order ABC when an auxiliary cement variable is defined on the surface.

The validation of the presented ABC is accomplished through the numerical examples in which both planar and curved surfaces are used as the truncation boundary. The superiority of the ABC to first order impedance boundary condition and its agreement with theoretical expectation are acknowledged from the experiments done. The results of these the numerical examples along with the details of the derivation and implementation steps and will be presented.

Time-Domain Finite Element Simulation of Periodic Structures

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Periodic structures have widely been utilized to control electromagnetic scattering and radiation for various engineering applications. Among the more common applications are frequency selective surfaces (FSS) and phased antenna arrays. A relative new application is the analysis and design of left-handed materials for which both the effective permittivity and permeability are negative. Periodic structures are usually analyzed in the frequency domain. However, there are two distinct advantages of analyzing periodic structures in the time domain. First, one can obtain a broadband frequency response in a single execution, and second, one can effectively simulate nonlinear devices or media. Unfortunately, there are two obstacles of applying the time-domain finite element method (FEM-TD) directly for the analysis of periodic structures. First, it is difficult to impose periodic boundary conditions in the time domain because a simple phase shift as it appears in the frequency-domain periodic boundary conditions translates into a time shift in the time domain. This time shift of the field at two parallel periodic surfaces requires future knowledge of the field at one of the two surfaces in order to enforce periodic boundary conditions. Second, it is difficult to formulate an accurate absorbing boundary condition (ABC) to be used to truncate the computational domain in the nonperiodic direction.

In this talk, we describe our recent work aimed at overcoming the two aforementioned obstacles. First, we introduce the concept of a transformed field as the working variable to formulate the FEM-TD solution of periodic problems. This transformed field permits a simple and direct enforcement of periodic boundary conditions in the time domain, although it complicates the governing partial differential equation significantly. Second, we formulate an accurate ABC, which is highly effective to absorb both the fundamental and the higher-order Floquet modes (whether propagating or evanescent, even if close to the grazing angle). Since this accurate ABC includes all the Floquet modes in its formulation, it is referred to here as the Floquet ABC (In practical applications, the inclusion of a few Floquet modes is sufficient to obtain very accurate solutions). Numerical examples are presented to validate the formulation both for scattering of plane waves from periodic structures as well as the radiation of phased-array antennas. These examples indicate that the Floquet ABC in conjunction with the transformed-field FEM-TD formulation is capable of producing highly accurate results even in the presence of higher-order propagating and/or evanescent Floquet modes.

An Extension of the Adaptive Absorbing Boundary Condition for the 3D Maxwell Equations

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We consider a large perfectly conducting structure on which is posed an inhomogeneous dielectric in a restricted zone. To solve electromagnetic scattering and radiation problems with this configuration, an efficient hybridization of higher order Finite Element Methods (FEM) and Boundary Integral Equations (BIE) based on an Adaptative Absorbing Boundary Condition (AABC) is then presented. It constists in truncating the computational domain by using an fictitious surface on which an AABC derived from a BIE is set. Unlike the standard FEM - BIE approach, this FEM-AABC coupling [1] [3] [2] constitutes a really powerful method for several reasons. The proposed method is free of interior resonance. It produces a purely sparse linear system matrix. Only the right-hand side of the linear system is changed from one iteration to another one. All the integrals involved in the formulation are not singular and therefore can be determined by quadrature formulas only. Finally, the radiation condition, set on an arbitrarily shaped fictitious surface which can be placed very close to the scatterer/radiator, is exactly satisfied as for standard FEM - BIE approach at convergence; and more importantly, the method produces a solution that converges to the true solution of the initial problem. However, despite these very attractive properties, the dispersion errors inherently attached to a FEM approximation of a wave phenomenon can become prohibitive: even if the fictitious surface can be set close to the scatterer/radiator, it must wrap up all the obstacle. In this extension, we propose a formulation, typically characteristic of an overlapping Domain Decomposition Method (DDM), enables to reduce the FEM discretization to a localized zone of the obstacle while the exterior computational domain, which can be a very large-sized part, is dealt with a BIE. There is a difficulty in the above extension comparatively to the original method as it was described in [2] and in [3]: the FEM determines currents only inside the dielectric part and the EFIE, used to deal with the exterior computational domain, is expressed by a hypersingular integral. This difficulty can be overcome using integration by parts to remove all the integrals that do not converge in the usual meaning. While keeping the main attractive properties of the FE-AABC approach, the FEM-BIE-AABC approach enables to reduce the computational domain attached to the FEM approximation and needs an iteration number lower than the standard FE-AABC approach. The required evaluation of boundary integrals in the calculation of the AABC and the resolution of the exterior BIE can be carried out using the multilevel fast multipole algorithm (MFMA). Unlike the standard FEM - BIE approach, it produces a purely sparse linear system matrix uncoupling to a purely dense linear system matrix. To demonstrate the great potential of this FEM-BIE-AABC approach, some tridimensional numerical results are given [4].

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Session 6 — Advances in Discrete Representations

Organizers

Botha, M. M.; Jin, J.-M.

Presentations

1. A novel class of differential form based interpolation operators for FEM
   Bluck, M. J.; Walker, S. P.

2. Decomposition of the mixed first-order, divergence-conforming function space on a tetrahedral mesh
   Botha, M. M.; Davidson, D. B.

3. Polygonal Finite Elements
   Euler, T.; Schuhmann, R.; Weiland, T.

4. Curvilinear Vector Finite Elements using Hierarchical Basis Functions
   Swartz, J. P.; Davidson, D. B.

5. A Singular Tetrahedral Finite Element for Sharp Metal Edges
   Webb, J. P.

A novel class of differential form based interpolation operators for FEM

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In this paper we discuss the construction of a novel class of interpolation operators for finite element methods. Interpolation operators play a central role in the discussion of convergence of such schemes. The operators defined in this paper are based on a differential form construction and offer advantages over current realizations of interpolation operators in the sense that they are defined as a scalar product. This results in operators which exhibit continuity and appropriate discrete hodge decompositions and in this sense are mimetic.
Basic properties such as the commuting de Rham diagram (CDD) are demonstrated, as is an adjoint form of CDD. It is shown that they also satisfy all of the requirements of a fortin operator which can be used to prove the convergence of certain classes of eigenvalue problem. Moreover, important properties such as discrete compactness are arrived at in a simple and general manner. Crucially, these results apply equally well to curvilinear problems as well as the strict polyhedral case.

Decomposition of the mixed first-order, divergence-conforming function space on a tetrahedral mesh

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The polynomial, mixed first-order, divergence-conforming function space on a tetrahedral element can be represented by a set of four basis functions, each associated with a face of the element [1]. In the case of a global divergence-conforming approximation over a mesh of elements, each basis function becomes associated with a global face in the mesh. This kind of representation is often used to model electromagnetic flux- and current densities in 3D. With such modeling, it would be beneficial to additionally split the global representation in terms of the solenoidal component (the null space of the divergence operator), and its complement. In the case of mixed first-order edge basis functions for curl-conforming representations, such a splitting, based on topological processing of the mesh, has been proposed to explicitly represent the gradient field (the null space of the curl operator) and its complement [2]. The latter splitting is referred to as *tree-cotree decomposition*.

Analogous to tree-cotree decomposition, we present here a splitting procedure for the divergence-conforming case [3]. We name this scheme the *leaves-coleaves decomposition* — it is also based on topological processing of the mesh and in fact utilizes the tree-cotree decomposition as a starting point. The presentation will be concluded with example results, showing some of the applications of the leaves-coleaves decomposition.

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POLYGONAL FINITE ELEMENTS

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In mesh-based electromagnetic simulations, triangular or quadrilateral elements are widely used for 2-dimensional domains. The Finite Integration Technique (FIT) [1] and the Finite Element Method (FEM) employing Whitney elements [2] transfer the geometrical properties of the gradient, curl, and divergence operator ($\operatorname{div}\operatorname{curl} = 0$ and $\operatorname{curl}\operatorname{grad} = 0$) onto the discrete operators. Now the question arises whether these properties can also be fulfilled using more general mesh elements. These mesh elements could allow for new meshing flexibility as the example of polygonal elements in Figure 1 shows. The full benefit of such general elements comes into play in 3-dimensional simulations, though, as the use of special prism [3] and pyramid elements [4] shows.

A general framework for the derivation of nodal, edge, and facial basis functions for arbitrarily shaped 2-dimensional polygonal elements will be presented. The commuting diagram properties responsible for the preservation of the geometrical properties will be outlined. These basis functions can be directly employed in the standard FEM or used to derive the material matrices within the framework of the FIT.

The numerical examples will show the wide application range for these elements in different electromagnetic formulations.

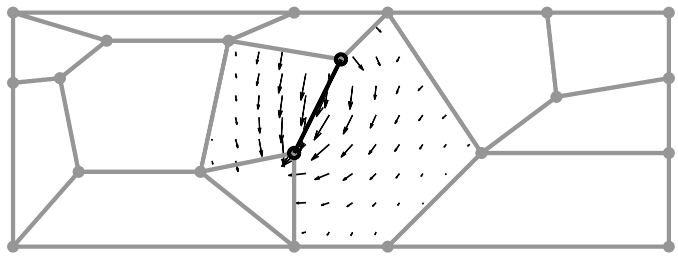

Figure 1: A small mesh consisting of arbitrary polygons. Shown in black is the vector plot of an edge basis function with its associated edge.

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Curvilinear Vector Finite Elements using Hierarchical Basis Functions

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As a numerical technique for the solution of Maxwell's equations of electromagnetism, the finite element method makes two simplifying approximations. Firstly, it aproximates the field variation by means of a set of basis functions and secondly, it approximates the geometry by discretizing it into a mesh of simple volumes. These volumes are usually rectilinear shapes such as tetrahedra, rectangular bricks or triangular prisms. Much work has been done on improving the field approximation by using higher order basis functions of both interpolatory and hierarchical types. While successive p-refinement is quite effective at reducing the numerical error for rectilinear geometries, the accuracy of the solution for curved geometries is limited by the error introduced by the geometrical approximation. This shortcoming can be addressed by using curvilinear elements [1] to approximate the geometry along curved surfaces.

An implementation of curvilinear elements employing a second order polynomial geometric mapping for the Webb [2] type hierarchical vector basis functions is presented. This geometrical mapping is used to transform the curvilinear element to a rectilinear reference element. The basis functions and their curls are derived for the reference element and used to compute the finite element matrices. The reduction in error provided by the improved geometrical approximation is illustrated by comparing rectilinear and curvilinear results obtained for eigenanalysis and scattering from spherical and cylindrical geometries.

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A Singular Tetrahedral Finite Element for Sharp Metal Edges

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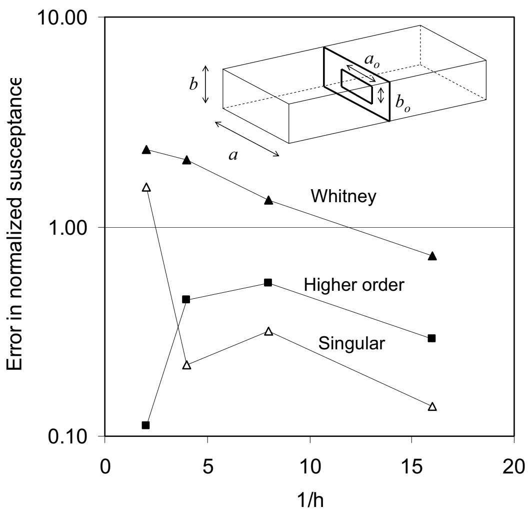
Conventional finite elements, based on polynomials, are not particularly effective at representing the singular electromagnetic field near sharp edges and corners (points). As a result, many degrees of freedom are generally devoted to such regions, usually by high mesh refinement, and this adds considerably to the cost of solution. An alternative is the use of special finite elements that can model the singular fields well. To date, efforts in this direction have been confined to two dimensions, i.e., the waveguide problem. Here the extension to 3D is considered.

The focus is on modeling metal edges, which are very common in microwave devices. Dielectric edges are not considered, nor are corners. Two new tetrahedral elements are proposed. *Line* elements have one edge along the sharp edge. *Point* elements have one vertex on the sharp edge. Both are needed to model the singularity. In the line element, a local coordinate system (ρ,σ,τ) is used which generalizes triangular polar coordinates [1]; roughly, (ρ,σ,τ) corresponds to (r,ϕ,z) of cylindrical coordinates. In the point element, the local system is $(\rho,\sigma,\tau,\upsilon)$; ρ measures distance from the singular vertex, and the other coordinates are similar to the barycentric coordinates of the non-singular vertices; $\sigma+\tau+\upsilon=1$.

To improve matrix conditioning, the new elements maintain a separation, as far as possible, between gradient and rotational basis functions [2]. Redundant vertex functions are included for the same reason. In both elements the gradient space consists of gradients of scalar functions of the form $\rho^{\nu}P$, where ν is a non-integer determined from the angle of the singular edge [3] and P is a polynomial in the local coordinates. The rotational space is built from functions of the form $\rho^{\nu}\left(P_\rho\mathbf{g}_\rho+P_\sigma\mathbf{g}_\sigma+P_\tau\mathbf{g}_\tau\right)$, where P_ρ,P_σ,P_τ are polynomials and $\mathbf{g}_\rho,\mathbf{g}_\sigma,\mathbf{g}_\tau$ are vectors parallel to the gradients of the respective coordinates. Both elements have 15 basis functions and are compatible with each other and with the standard Whitney edge element (6 basis functions).

The new elements were used to determine the susceptance of a metal iris in a rectangular waveguide. The graph shows error (with respect to a highly converged solution) versus the inverse of the maximum element size, as the initial mesh is uniformly refined. The "Whitney" curve was found using entirely Whitney elements. Replacing the Whitney elements touching the sharp edges with conventional elements of one order higher gives the "Higher order" curve; replacing them with the new, singular elements gives the "Singular" curve.



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Session 7 — FEM Applications: Antennas and Materials. Part 2

Organizers

Kempel, L.; Meyer, F. J. C.

Presentations

1. Eigen value Estimation of Finite Element-Boundary Integral Matrix Systems and their Application to FSS Design
   Fischer, B. E.; Volakis, J. L.;Yagle, A. E.

2. Inhomogeneous Material Loading for Microstrip Leaky-wave Antennas
   Jiang, H.; Pasala, K.; Kempel, L.

3. Development of Time-Domain Finite Element Method for Analysis of Broadband Antennas and Arrays
   Lou, Z.; Jin, J.-M.

4. Finite Element Method for Periodic Structures
   Yarga, S.; Sertel, K.; Volakis, J. L.

Eigen value Estimation of Finite Element-Boundary Integral Matrix Systems and their Application to FSS Design

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Over the years, numerical methods, and more specifically finite element techniques have been used for modeling guided wave structures to identify their propagation characteristics. In the general sense, use of numerical techniques has to a great degree focused on solving complex and/or large scale problems involving a combination of material and composite structures. In this paper, we take a closer look at the general-purpose Finite Element-Boundary Integral (FE-BI) methods, and carry out an eigen-decomposition aimed at identifying the propagating modes without a need to directly solve the exact FE-BI system. The resulting eigenvalues are associated with propagating modes within frequency selective surfaces (FSSs) and volumes (FSVs), and are used for designing such structures by modifying the periodic cells and possibly the dielectric constant of the substrate.

Key to accomplishing the proposed approximate eigen-decomposition is the Green's function expansion

$$e^{-jkR}/R \approx e^{-jk_0R}\left(1 - j\left(k - k_0\right)R - 0.5\left(k - k_0\right)^2 R^2 + \ldots\right)/R \cdot \tag{1}$$

We will use this approximation to obtain a matrix approximation that explicitly identifies the frequency dependence. Specifically, by retaining just the first term of **Error! Reference source not found.**, the FE-BI system can be re-written as

$$\left\{\mathbf{A}^{(1)}\left(k_0\right)+\tilde{k}^2\mathbf{A}^{(2)}\left(k_0\right)+\tilde{k}^2\mathbf{G}^{(1)}\left(k_0\right)+\mathbf{G}^{(2)}\left(k_0\right)\right\}\mathbf{e}\left(k\right)\approx\mathbf{f}\left(k\right), \tag{2}$$

where $\tilde{k} = k/k_0$ is the normalized wavenumber, $\mathbf{A}^{(1)}$ and $\mathbf{A}^{(2)}$ refer to the usual two terms of the finite element functional, and $\mathbf{G}^{(1)}$ and $\mathbf{G}^{(2)}$ correspond to the matrices resulting from the first two terms of the expansion (1). In this manner the frequency dependence is totally contained in the coefficients of the sub-matrices in (2). The identification of the eigenvalues (for the matrix multiplying the eigenvector **e**(k)) can proceed by introducing the factorization

$$\mathbf{X\Lambda X}^{-1}=\left\{\mathbf{A}^{(2)}\left(k_0\right)+\mathbf{G}^{(1)}\left(k_0\right)\right\}^{-1}\left\{\mathbf{A}^{(1)}\left(k_0\right)+\mathbf{G}^{(2)}\left(k_0\right)\right\}, \tag{3}$$

where $\mathbf{\Lambda} = \mathrm{diag}\left(\lambda_i\right)$. Substituting this into (2) yields the solution

$$\begin{aligned}\mathbf{e}\left(k\right)&\approx\left(\tilde{k}^2\mathbf{I}+\mathbf{X\Lambda X}^{-1}\right)^{-1}\left\{\mathbf{A}^{(2)}\left(k_0\right)+\mathbf{G}^{(1)}\left(k_0\right)\right\}^{-1}\mathbf{f}\left(k\right),\\&=\mathbf{X\Lambda}_k^{-1}\mathbf{X}^{-1}\left\{\mathbf{A}^{(2)}\left(k_0\right)+\mathbf{G}^{(1)}\left(k_0\right)\right\}^{-1}\mathbf{f}\left(k\right)\end{aligned} \tag{4}$$

where $\mathbf{\Lambda}_k = \mathrm{diag}\left(\lambda_i + \tilde{k}^2\right)$. The resonances/propagating wavenumbers for the subject FE-BI system are then given by (approximately)

$$\tilde{k}_i = \sqrt{-\mathrm{Re}\left\{\lambda_i\right\}} \cdot \tag{5}$$

Correlation of these wavenumbers with the structural properties allows for designing spatial filters, optical mirrors and reflectors specifically designed to best respond at pre-specified frequencies. At the meeting, we will show how *apriori* knowledge of the eigenvalues can be used to design for patch antennas on textured substrates and FSSs/FSVs or other periodic structures consisting of metallic and dielectric inclusions. Above all, the eigenmodes developed under this construct provide a demonstrably unique insight into the resonant behavior of an arbitrary volumetric cavity. Such insight suggests a capability for powerful new design and optimization tools for antennas.

Inhomogeneous Material Loading for Microstrip Leaky-wave Antennas

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Microstrip leaky-wave antennas are very thin (typically less than several millimeters for the antenna itself) and offer the potential for operation over a wide bandwidth as compared to its resonant cousin, the microstrip patch antenna. The theory of a full-width microstrip leaky-wave (FWLW) antenna is well developed by a variety of authors including Oliner and his students [1-2] and Chang *et al.* [3]. A related aperture, the half-width leaky-wave (HWLW) antenna, has been the subject of considerable recent attention due to its reduced lateral dimension (to aid arraying) [4-6]. It has been found that the functional behavior of the FWLW and HWLW antennas are similar and can be related rather simply. Hence, it is sufficient to examine one or the other as convenient.

All of the previous work involved a homogeneous substrate. It can be shown that the properties of a microstrip leaky-wave antenna depend on the width of the microstrip and the substrate (dielectric constant and thickness). Effects of strip width have been investigated [7]; however, not the effect of material variation. In this work, the impact of material changes on the radiation properties of the antenna is investigated for both lateral variations and longitudinal variations. The hybrid finite element-boundary integral (FE-BI) method is ideally suited for this application owing to the efficiency of the solution representation (as compared to a volumetric integral equation) and high fidelity geometric representation (as compared to conventional finite difference-time domain methods). Both near-field metrics (e.g. VSWR, driving-point impedance, etc.) and far-field metrics (e.g. gain and pattern) will be discussed.

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Development of Time-Domain Finite Element Method for Analysis of Broadband Antennas and Arrays

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Broadband antennas have been increasingly used in modern commercial and military communication and surveillance systems. Typical broadband antennas have very complex structures to achieve a desired bandwidth with a compact design. More advanced designs make use of exotic electronic materials, such as artificial magnetic conductors and electromagnetic bandgap materials, to further enhance the performance. Because of the complex structures and materials employed and their stringent performance specifications, the design of broadband antennas through purely experimental means has become very time consuming and expensive. To reduce turn-around times and development costs, one must resort to computer aided analysis and design (CAD) tools to systematically iterate through, and thereby optimize, designs.

In this talk, we propose the use of the time-domain finite element method (TDFEM), as a promising CAD technique for modeling and simulation of complex broadband antenna designs. The TDFEM has inherent advantages for modeling complex structures and material compositions because of the use of unstructured tetrahedral meshes and for modeling nonlinear devices and materials because it works directly in the time domain. Using an implicit scheme such as the Newmark-beta method, the TDFEM can deal with tiny elements (with dimension on the order of a few thousandths of a wavelength) without any severe penalty on the size of the time step and on the time-marching stability. There are three main challenges for the TDFEM that have to be overcome to make it a powerful, efficient, and accurate method for the simulation of broadband antennas. The first one is the mesh truncation. Whereas the implementation of the simple time-domain absorbing boundary conditions (ABC) is rather straightforward, its performance is limited due to significant reflection from near-grazing incidence. A better choice is to use perfectly matched layers (PML) for mesh truncation; however, unlike the case for the FDTD and frequency-domain FEM, the PML implementation in the TDFEM is very complicated. The second challenge is the accurate modeling of the antenna feed. A robust scheme for modeling antenna feeds is critical to the design of advanced antenna systems such as multi-functional antenna arrays. While simplified feed models permit the calculation of reasonably accurate radiation patterns, they fail for all but the simplest systems when input impedances and mutual coupling (S-parameters) are to be accurately predicted. The third challenge is the need to solve a large sparse matrix in each time step regardless the time-marching scheme used (either implicit or explicit), which has made the method very inefficient for large-scale simulations. Novel numerical schemes are required to either solve such a large sparse matrix very efficiently or to avoid such a solution completely. In this talk, we will discuss the solutions to these problems and present numerical examples to demonstrate the capability of the proposed TDFEM.

FINITE ELEMENT METHOD FOR PERIODIC STRUCTURES

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Theoretical developments in periodic material modeling and experimental demonstrations of extraordinary phenomena exhibited by engineered metamaterials have broadened the perspective of designing multipurpose integrated antennas as well as miniature microwave devices (*IEEE Trans. on Ant. and Prop.*, Special Issue on Metamaterials, vol. 51, Oct. 2003). Among these designs, the degenerate band edge crystals (DBEs), shown to support slow propagation modes (Mumcu, K. Sertel, and J. L. Volakis, 2005 *IEEE APS Conf.*, July 2005-134), are well suited for high sensitivity and high directivity miniature antennas due to the accumulation of the incident electromagnetic energy. These DBEs consisting of alternating layers of misaligned anisotropic slabs may be constructed using available low-loss anisotropic materials such as rutile (TiO_2) crystals. A more flexible design of such anisotropic layers uses metallic inclusions embedded within a homogeneous host medium (H. Kubo, T. Iribe, A. Sanada, and I. Awai, *Asia-Pacific Mic. Conf. Proc.*, 588–1591, 2002). Nevertheless, homogenization models for such thin layers are simply not accurate due to the surface depolarization effect. The other alternative in designing the required DBE characteristics is to model the geometry exactly within a finite element (FE) framework. Although analytical methods already exist for generating band diagrams for simple periodic media, for metamaterials composed of high contrast materials and/or metallic inclusions, standard plane-wave expansion or mode matching methods fail to converge due to the abrupt changes within the unit cell geometry. In this paper, we demonstrate the use of a triply periodic finite element method formulation for designing the required band structure of a DBE crystal consisting of metallic inclusions.

Analysis of periodic structures is solely based on the assumption that field distribution within the medium is also periodic (Bloch-Floquet theorem). The resulting modes within the crystal (Bloch modes), have a periodic field envelope in each unit cell, differing only in their phase. The latter is associated with the propagation constant of the Bloch mode. As the field envelope is periodic, one can reduce the computational domain to a single unit. The standard FE functional reduces into an eigenvalue problem after imposing the periodic boundary conditions across the corresponding faces of the unit cell (assuming a cubic lattice). The periodic boundary conditions also eliminate the mesh truncation requirements since all six faces of the computational domain are linked via the periodic boundary conditions. Given the Bloch wave number, the solution of the eigensystem produces the allowable frequencies for the modes of the discretized unit cell geometry. Then, a sweep of the wave numbers leads to the full band structure of the periodic design. Furthermore, the field distributions of these allowable modes can be obtained from the computed eigenvectors of the system. The full FE nature of this approach allows for an efficient solution procedure that can be used within a design loop. We will demonstrate the design of the DBE using this triply periodic FE solver along with the implementation details such as the hexahedral vector finite elements and the geometrical details.

Session 8 — Special Session in Honour of P. P. Silvester

Organizers

Ferrari, R. L.

Presentations

1. Papers by P. P. Silvester applying integral equation methods – a historical review
   Ferrari, R. L.

2. The contribution of P.P. Silvester to Wave Electromagnetics
   Pelosi, G.; Selleri, S.

3. The early contributions of P. P. Silvester to the finite element solution of saturable magnetic field problems
   Webb, J. P.

Note that a complete bibliography of the published papers and books of P. P. Silvester is appended to these abstracts. The bibliography was compiled by R. L. Ferrari.

Papers by P.P.Silvester applying integral equation methods – a historical review

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In electromagnetics numerical analysis the relative significance of integral-equation (IE) methods, with their reduced dimensions and ability to deal with open boundaries, compared with other approaches discretizing a whole working-space has never been clear. From the earliest days P.P.Silvester (PPS) was aware of this dichotomy and kept his options open. We can classify as an 'integral method' one characterized by the appearance of a Green function $G(P,Q)$ which, singular at $P = Q$, is discretized and applied numerically over some boundary using a Galerkin /variational procedure [145]. We also include the accurate postprocessing of volume-FEM results through application of Green's vector theorem. Then some 25% of the PPS lifelong journal output falls into these categories. This output will be briefly reviewed in the presentation.

In 1967 using image theory PPS devised [9] a Green function for the quasistatic electric field of a microstrip transmission line fabricated as a perfectly conducting (PEC) ribbon overlaying an infinite dielectric sheet and PEC ground plane. He applied the Green function to its discretized boundary problem which was solved using Galerkins's method to provide computer estimates of microstrip line capacitance. In 1972-3 five more papers [35-38,51], plus a program description [48], with P.Benedek and one with A. Gopinath [46] expanded this work to the modelling of various microstrip configurations. A 1995 revisitation [143] related to multiconductor lines .

In the same early period, with A. Cermak and others, PPS initiated a 'boundary relaxation' procedure, connecting a finite differences (FDM) region to an external unbounded one represented by an integral equation, in static and low frequency electromagnetic problems [10,11,20-22]. By the 1970's PPS had replaced FDM by the finite element method (FEM) and, with M.S.Hsieh, [30] solved the 2D open transmission line problem by means of a hybrid FE/IE approach, also providing a detailed discourse [32] on the 'projective' (*i.e.* weighted residual/Galerkin) solution of IE problems.

In the 1970's PPS also became interested in the numerical solution of the standard Hallen and Pocklington wire-antennas IE's. He noted that the contemporary and popular point-matching techniques, effectively using a delta-function weighting, were not variational and therefore inferior to projective Galerkin methods. With principle coauthors K.K. Chan [34,43,52,57] and M.A.Hassan [64,69] six papers in 1972-76 ensued.

Then there was inactivity with respect to IE's until 1985, when PPS briefly observed [94] that the accuracy of quantities, *e.g.* field gradient or forces, derived by differentiation from standard volume FEM result could be considerably improved by means of algorithms based upon an application of Green's integral theorem. There followed some nine papers[111,116,118,125,127,128,135,147,148] and then one posthumously [155] on this theme, six of them coauthored with D. Omeragic .

Reference numbers here refer to the PPS bibliography addendum to this volume

The contribution of P.P. Silvester to Wave Electromagnetics

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The Finite Element Method (FEM) is a variational technique solving differential equations. The variational principle, indeed, starts by defining an expression, or functional, of a field, be it an electromagnetic field, a pressure field, a stress or torsion field solution of the differential equation. This functional is known to have a stationary point (be it a maximum or a minimum), when the field attains the values which are solution of the differential equations. Usually such a functional is related to the energy stored in the domain and the solution is the field minimizing such energy.

The FEM was introduced firstly by Courant in 1942 in his paper *Variational methods for the solution of problems of equilibrium and vibration.* This work contained an appendix in which Lord Rayleigh's potential was solved for via a variational approach over a domain subdivided into triangles, which Courant himself called *elements*.

The first true application of Finite Elements to electromagnetic wave equations, and hence to high frequency electromagnetics, dates back to 1968-1969. P.P. Silvester published in *Alta Frequenza*, the Italian national High Frequency electromagnetics journal, a paper *Finite-element solution of homogeneous waveguide problems* at the beginning of 1969. In this fundamental paper P.P. Silvester solved the problem of mode determination in a homogeneous perfect-electric-conductor bounded waveguide.

His effort in presenting Finite Elements in a comprehensive and coherent fashion is universally recognized and may be dated back to his derivation of universal matrices, leading to the very well known *Finite Elements for Electrical Engineers* (1983) co-authored with Dr Ferrari, which is still a textbook of choice for studying Finite Elements in wave electromagnetics.

In this contribution the solution presented by P.P. Silvester in his 1969 paper will be also presented, and its impact on the electromagnetic community analyzed.

The early contributions of P. P. Silvester to the finite element solution of saturable magnetic field problems

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Between 1970 and 1973 Silvester co-authored 4 papers, [23][27][29][47], that established a new and important branch of computational electromagnetics: the finite element solution of saturable magnetic field problems. Between them the papers have received over 280 citations; the first in the sequence, with 131 citations, is the second most referenced of Silvester's papers. As a result of this early work, the finite element method emerged as the primary tool for the simulation of magnetic fields in power-frequency devices, such as motors, generators, transformers and actuators.

In 1970, the finite element method was very new to electrical engineers. Though the applications in mechanical engineering were quite developed, only a few papers on electrical topics had so far appeared – mostly authored by Silvester himself [12-15]. It was known that first-order triangular elements could be used to solve the Poisson equation in general, but only in [23] was the idea introduced of using them to solve the specific equation of 2D magnetostatics:

$$\nabla \cdot \nu \nabla A_z = -J_z \tag{1}$$

For many magnetic devices, a cross-sectional (xy) analysis is sufficient, with a z-directed current density, J_z. Then only one component of the magnetic vector potential is needed, A_z, and the problem is scalar. However, it is nonlinear, since the reluctivity, ν, is a function of the magnetic flux density (and therefore of A_z). In [23] it was shown how this nonlinear problem could be written in variational form using the energy density, and the stationary point found numerically by the Newton-Raphson method. A simple transformer geometry was analyzed. In the following three papers, the same approach was used to predict the performance of rotating machines.

Before this, the prevailing method for the analysis of rotating machines had been the finite difference method (FDM). We are fortunate to have, appended to each of the first three papers, a discussion of the paper by other experts, several of whom had developed FDMs and were quite hostile to the new approach. This allows us to see the contribution of this early work in historical context. The assumption in 1970 was that, to model an electrical machine adequately, you needed many (around 10^4) finite difference grid points, and the emphasis of the research was on the speedy solution of large, sparse, highly-structured, nonlinear systems of simultaneous equations. Silvester's approach was quite different. By using finite elements, a machine could be modeled with a very small number of degrees of freedom – 151 in [27] – sufficiently well to give terminal parameters accurate to a few percent. The accuracy could be achieved with so few degrees of freedom, of course, both by virtue of the variational approach and because it was possible to mesh the complex geometry with relatively few triangles, compared to the number of grid points needed by the FDM. In the early 1970s, this approach was revolutionary.

Reference numbers refer to the bibliography of the works of P. P. Silvester, which appears as an addendum to this volume.

Peter P. Silvester (1935-96)
A Complete Bibliography of Published Papers and Books

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Session 9 — Time domain FEM. Part 1

Organizers

Hesthaven, J. S., Rylander, T.

Presentations

1. Applying the Cement Technique to Non-Conformal Finite Element Time Domain Method for Electromagnetic Problems in R3
   Venkatarayalu, N.; Gan, Y. B.; Lee, J.-F.

2. A Higher Order Discontinuous Galerkin Time Domain Method for Beam Dynamics Simulations in Linear Accelerators
   Gjonaj, E.; Weiland, T.

3. Comparison of Time Domain FEM Formulations
   Marais, N.; Davidson, D. B.

4. Use of Hanging Variables in the Time Domain for Adaptive Refinement and Subgridding
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5. Discontinuous Galerkin Time Domain FEM for Solving Maxwell Equations
   Wong, M.-F.; Wiart, J.

Applying the Cement Technique to Non-Conformal Finite Element Time Domain Method for Electromagnetic Problems in $\mathcal{R}^3$

Neelakantam Venkatarayalu   Yeow Beng Gan   Jin-Fa Lee

The finite element time domain method provides an attractive alternative to the finite difference time domain (FDTD) method for solving time-varying Maxwell equations in 3D. Particularly, in situations where accurate geometrical modeling is needed and/or fine features are involved within the problem domain. However, in many practical applications, the performance of the standard FETD algorithm is simply too slow compared to the explicit FDTD method. This concern sparks significant research interests in hybridize the FETD and the FDTD in an attempt to retain the benefits and not suffer the pitfalls of each method. Worth mentioning is also the Discontinuous Galerkin FETD algorithm, which essentially diagonalizes the finite element system matrix equation, and marching the solution in time explicitly. Combining with local higher order basis functions, the result is a very efficient time domain solver for Maxwell equations in 3D.

In this work, we take a different but yet similar approach to the DG to formulate the FETD algorithm for non-conforming finite element meshes for different domains. We took our cue from our recent works in domain decompositions, and specifically the cement technique introduced previously. We should remark here that we have implemented the standard mortar-type interface conditions to join dis-similar finite element meshes across different domain blocks, and the preliminary results indicate the standard Dirichlet-Dirichlet transmission condition employed in the mortar technqiue suffers greatly the internal resonances in the time domain marching algorithm. To mitigate the difficulty, we introduce two sets of cement variables and *glue* dis-similar domains together through a Galerkin/variational setting. The details of the formulation and its numerical performances will be discussed in full during the talk.

A Higher Order Discontinuous Galerkin Time Domain Method for Beam Dynamics Simulations in Linear Accelerators

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The simulation of collective effects in particle accelerators is a major numerical challenge. This is related to the broad frequency spectrum of the space charge fields involved in the relativistic and ultrarelativistic accelerating regime. Traditional time domain methods fail to resolve, both, the current and field distribution within the nm-range of short electron bunches. In particular, when used for Particle-In-Cell simulations, these methods produce excessive numerical noise as resulting from the high dispersion error.

In this work, a higher order, energy conserving Discontinuous Galerkin Time Domain method for this type of applications will be presented. The idea motivates, primarily from the superior behaviour of dispersion error for higher order discretizations. Additionally, the method is implemented on octree-based grids, thus, allowing for highly local and adaptive mesh refinement. We will discuss issues related to the stability, energy and charge conservation as well as to the dispersion properties of the method. For wake field computations, directional splitting time integration techniques will be presented.

Comparison of Time Domain FEM Formulations

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Frequency domain Microwave FEM is almost completely dominated by the standard curl-curl vector-wave equation formulation. For Finite Elements Time Domain (FETD) there are several available formulations [1], including ones based on the standard curl-curl formulation. Apart from the FEM formulation, di erent choices of time integration method may also be made.

The choices present di erent tradeo s. Certain combinations of of formulation and time integration can be utilised to make explicit (i.e. they require no matrix solution) methods similar to the FDTD [2], whereas other choices make unconditionally stable methods, that have no inherent time-step size limitation analogous to the Courant limit in FDTD [3, §10.7].

In practice it is not clear cut which set of tradeo s would be most bene cial. One gure of merit to be considered is the computational efficiency achieved solving a given problem geometry to a given accuracy. The relative ordering of several methods given this criteria would still vary depending on the particular problem geometry and desired solution accuracy.

In this presentation, a FETD formulation based on the vector-wave equation and one based on the two rst-order coupled Maxwell equations are compared. Several geometries and various accuracy criteria are considered. The formulations are also compared on less quantitative criteria, such as the ease of implementation and their suitability for solving di erent classes of problem.

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Use of Hanging Variables in the Time Domain
for Adaptive Refinement and Subgridding

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The concept of hanging variables has been around for the past five years, and it has been used for refinement in frequency domain finite element methods. In the work presented here, hanging variables are extended to the finite element time domain (FETD) method for adaptive mesh refinement and the finite difference time domain (FDTD) method for subgridding. In the FETD method, a short time run simulation is done to generate a solution for refinement. The simulation time is selected such that the energy reaches every location in the initial mesh. Then, a discrete Fourier transform (DFT) is employed at specified frequencies to generate frequency domain solution. After the DFT, an *a posteriori* analysis is performed in each element for all specified frequencies. Refinement is then done through the use of nested meshes in those elements where the error estimate indicates that refinement is necessary. After the initial refinement, the resulting grid will contain fine elements bordering coarse elements. The hanging variable approach will allow the tangential field to be continuous at the interface between the coarse and fine elements. The process continues until the specified criterion is satisfied. The approach presented here has been implemented for the two-dimensional case, and results will be presented to demonstrate the capabilities of such an approach. The use of hanging variables has also been adapted for subgridding in FDTD. It is well recognized that most subgridding approaches are unstable, typically suffering late time instability. The approach we have taken is to use to use a hybrid FETD/FDTD method where the subgridding is done within the FETD portion of the grid. The hanging variables are used to maintain the correct continuity conditions at the subgrid interface. Also, because the formulation is based on FETD, the stability of the approach can be guaranteed. Although finite elements is used in performing the subgridding, the implementation of the final algorithm can be done in much the same way as traditional FDTD where a stencil is defined with appropriate coefficients. In 2D, the subgridding scheme is explicit; however, in 3D, the scheme is implicit. Because of the sparsity of the resulting matrix, there is no increase in computational complexity compared to FDTD due to the implicit nature of the subgridding. Results for 2-D and 3-D will be presented.

Discontinuous Galerkin Time Domain FEM for Solving Maxwell equations

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Although the Finite Difference Time Domain (FDTD) is dominating the modeling of electromagnetic systems involving heterogeneous media, other approaches are investigated to achieve better accuracy and efficiency for solving ever increasing complex electromagnetic problems. The underlying regular orthogonal grid of the FDTD; which explains its simplicity and performance, limits its ability to tackle efficiently arbitrarily geometries and fine details. Amongst possible approaches are locally modified FDTD, sub-gridding techniques. Finite Element Time Domain methods are naturally able to conform its mesh to the geometry but the algorithms are generally implicit requiring a matrix solving at each time step. Mass-lumping technique is not straightforward for general meshes like tetrahedral ones. Recently, discontinuous Galerkin methods have been proposed in solving Maxwell equations in time domain. It uses local basis functions which can be high order ones within each element to decompose the electromagnetic fields. The fields within elements are connected through the centered mean approximation of the surface integrals of the finite element formulation. In finite element framework, this is quite unusual as generally the continuity through elements are features of the basis functions. The time step algorithm uses the classical leap-frog scheme. This technique presents many interesting features. The formulation applies to orthogonal hexahedral meshes and to tetrahedral meshes as well. For the first kind of meshes, similarities with the Transmission Line Matrix (TLM) using symmetrical condensed node can be found. For tetrahedral meshes, it can be seen as a way to make the algorithm explicit as only small size matrices need to be solved. The algorithm verifies the conservation of a discrete energy for different order of the basis functions whose property can be verified also when using non-conforming meshes. Despite a higher number of degrees of freedom, this technique is promising as a flexible way to tackle explicit and stable schemes and further enabling combinations of non-conforming meshes and different order basis functions.

Session 10 — FEM Applications: Waveguides, Components, Active Devices, Lumped Elements

Organizers

Mittra, R.; Selleri, S.

Presentations

1. Simulating the Performance of Electronic Packages and Printed Circuit Boards
   Bracken, J. E.; Polstyanko, S.; Cendes, Z. J.

2. Implementation of Lumped Elements and Fast Frequency Sweep into the Vector Finite Element code for RF Electromagnetic Field Simulation
   Grigoriev, A. D.; Salimov, R. V.; Tikhonov, R. I.

3. A New FEM/BEM Domain Segmentation Approach For Solving High Diversity Electromagnetic Problems
   Guarnieri, G.; Pelosi, G.; Rossi, L.; Selleri, S.

4. Modeling Wave Guiding Structures at Very Low Frequencies
   Lee, S.-C.; Lee, J.-F.

5. Model Order Reduction of Finite Element Models of Electromagnetic Structures with Embedded Frequency-Dependent Multiports
   Wu, H.; Cangellaris, A. C.

Simulating the Performance of Electronic Packages and Printed Circuit Boards

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Modern CPU clock frequencies are advancing toward 4 GHz. At the system level, front-side bus speeds are approaching 1066 MHz and wired networks operating at 10 Gb/s are available. System designers would like to accurately evaluate the electromagnetic interference produced by these high-speed signals. We present a specialized 2D finite element method solver to compute the full-wave electrical behavior of such complex boards and packages.

Consider two parallel metal power/ground planes parallel to the (x, y) plane, separated by a dielectric layer with thickness *d*. This structure can support a TEM mode as well as higher order TE and TM modes. However, if the metal planes are electrically close, higher order modes can be neglected due to their high cut off frequency and then only a TEM mode has to be considered. This mode is characterized by a *z*-directed electric field and *x-y* directed magnetic field, both independent of *z,* and can be obtained by solving wave equations. In order to handle arbitrary-shaped metal planes, we employ a triangular discretization and first-order nodal basis functions.

A digital board or package will have many signal traces passing between the voltage supply planes. When a trace is present, the total electric field is no longer vertical. We can circumvent this difficulty by using a modal decomposition technique to decouple the transmission line modes from the plane modes. The transmission line modes are modeled with analytic formulas and the supply planes using 2D FEM. At the endpoints of the traces, vias are used to connect traces to power/ground planes or external circuitry. At these points traces and planes are coupled and energy is transferred between modes.

The Xilinx ML481 Test Board and Virtex-4 LX60 FF1148 Package is shown in Figure 1. Comparison of simulated and measured results at various "spyholes" in the circuit show good agreement.

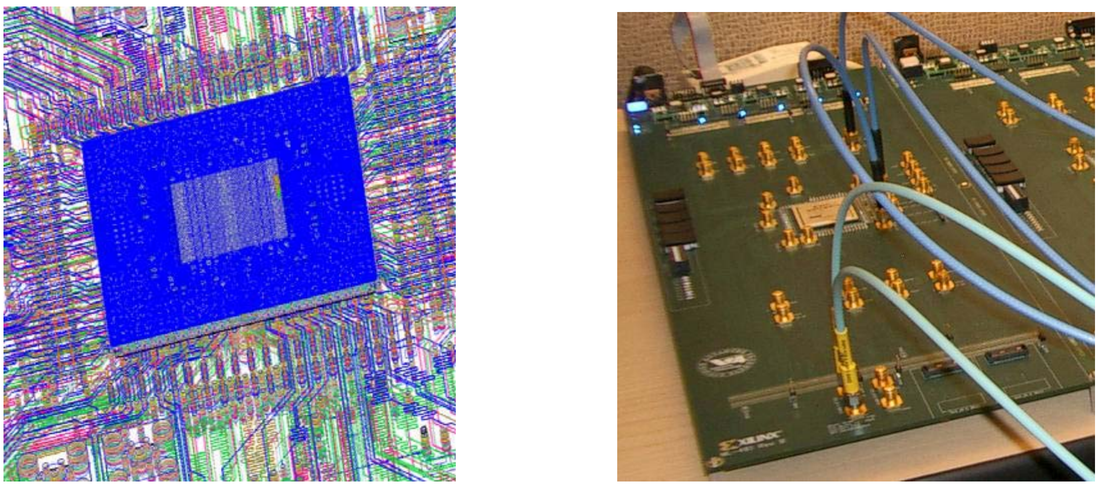

Figure 1. Simulation mirrors measurement.

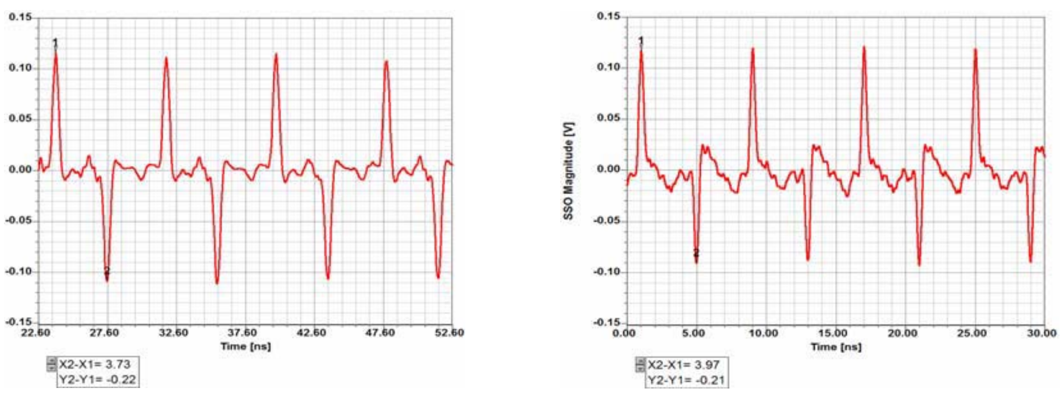

Figure 2. SSO simulation (left) versus measurement (right) at spyhole L34.

Implementation of Lumped Elements and Fast Frequency Sweep into the Vector Finite Element code for RF Electromagnetic Field Simulation

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Efficiency and functionality of a frequency domain vector finite element method (VFEM) for RF electromagnetic field simulation can be substantially increased by implementation of lumped elements (LE) and fast frequency sweep (FFS) technique. Both items ought to be compatible with each other in order to achieve maximum efficiency.

Usually LE is defined as a series or parallel *R*, *L*, *C* circuit. Impedance or admittance of such a circuit is a rational function of frequency. So, FE global matrix and right hand vector (RHV) elements associated with LE contain members with negative frequency power, which is inconsistent with FFS. Expansion of such elements into a Taylor series around the central frequency results in a row with infinite number of frequency moments. Truncation of this row causes corresponding loss of accuracy. In this work it is shown that multiplication of matrix rows containing entries with negative wave number power by an appropriate polynomial of complex wave number allows obtaining the global matrix with entries of minimum zero and maximum 5-th degree of complex wave number and RHV of maximum 4-th degree. Such matrix and vector can be exactly expanded in a Tailor series with finite number of members (moments) necessary for FFS implementation.

Among numerous FFS algorithms described up to date, the well conditioned asymptotic waveform evaluation (WCAWE) [1] was chosen for implementation due to its robustness and wide frequency range covering by one frequency point solution. The critical point of this algorithm is the optimal choice of a number of unknown vector moments Q. Inadequate choice results in inaccurate frequency response or in extra solution time. Special algorithm was developed for automatic optimal choice of Q, based on moment norm calculation and monitoring its changes from q-th moment to moment q+1. The process of orthogonal basis vector evaluation stops when the difference between norms of successive vectors is less then the prescribed value.

The described algorithms were implemented in a VFEM computer code for RF electromagnetic field simulation and circuit parameters evaluation. The code has advanced graphic user interface and calculates near field pattern, scattering matrix of the circuit, radiation pattern, directivity, gain and efficiency of microwave antennas. The algorithm was tested on various waveguide and antennas problems and showed high effectiveness. A number of simulation examples of concerning waveguides, microstrip lines and antennas problems are presented.

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A New FEM/BEM Domain Segmentation Approach For Solving High Diversity Electromagnetic Problems

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Nowadays, the increasing computer performances permits to deal with very high complexity electromagnetic problems, like semiconductor devices, their interfacing and packaging with Finite Elements (FE) or other numerical techniques. On the other hand the use of FE in synthesis tools, requiring many consecutive analyses, is still dramatically time-consuming. The numerical complexity of an electromagnetic FEM simulation is given mainly by the number of degree of freedom (dof) of the problem itself, which is tightly bound to the geometrical and physical characteristics of the domain. Therefore there is the need to find general techniques to minimize the total dof number, usually by means of problem- specific methods, as for example, the segmentation method for waveguide device synthesis problem [1,2].

Electromagnetic problems may exhibit a large domain whose geometry is generally simple, except for a small sub-domain in where a very high complexity or simple tiny structures whose position/shapes are to be optimized are present. In this case the computational burden is given by the total problem size, and the number of dof is necessarily high, and by any optimization or special equation to be performed or enforced in the small complex sub-domain.

Here we suggest to use a generalization of the Huygens principle to decouple the global problem into two distinct sub-problems: a Complex Domain (CD) problem spanning the region of the domain in which optimization or special equations are to be applied and a Large Domain (LD) problem characterized by a simple structure but a very large size.

The CD is, therefore, a reduced size problem with a fictitious boundary on which are tangential electric and magnetic field continuity conditions are to be enforced. This can be done by resorting to the Green function of the LD on the fictitious boundary decoupling the two domains and original conditions on physical boundaries of the domain. Being the Green function of the external domain unknown, in general case, a pre-processing analysis is required in order to compute it with the desired accuracy. This process, which can be carried out using FE, is obviously time-expensive, but is required only once for any given problem.

Indeed the proposed technique permits to "project" the LD and its boundary conditions on the fictitious boundary of the CD, obtaining a reduced size problem on which simulations or synthesis will be performed with great speed advantages with respect to the standard full domain FE solution.

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Modeling Wave Guiding Structures at Very Low Frequencies

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The analysis of two-dimensional wave guiding structures is of significant importance in electromagnetics community. It provides impedance information of the transmission lines, the modal configuration of energy propagation, and it is becoming more popular that the two dimensional eigen analysis is an integral part of studying real life three-dimensional electromagnetic problems. Efforts have been carried out by many authors previously using various frequency domain and time domain numerical electromagnetic techniques. Among them, the finite element method (FEM) is the most suitable and versatile in handling arbitrary material properties and geometric shapes.

However, the FEM implemented with the popular edge elements may encounter low frequency instability. Namely, when the operating frequency becomes very low, the solution process may break down and fail to provide any meaningful solutions. Furthermore, the same deficiency can take place when the element size becomes much smaller than the wavelength. To remedy this difficulty, a tree-cotree splitting of the edge elements is proposed and implemented. By the splitting, the edge elements are decomposed into either pure gradient or their complements, which leads to an approximate realization of the Helmholtz decomposition.

To further improve the efficiency of the solution process, a constraint equation is also proposed and implemented in this work. It can be easily shown that the finite element system of equation also supports non-physical eigen modes. Although they can be easily identified, the occurrence of those modes degrades the numerical efficiency and stability of the finite element process. In this regard, a constraint equation based on the orthogonal property of the physical modes is derived and imposed in an eigen solver to suppress the non-physical modes.

The stability of the proposed method at low frequencies is verified by a sample example. The eigen modes for power lines buried underground are obtained by the current method. Even though the smallest element size in this example is smaller than 1/10,000 of the wavelength, the convergence of the solution could be obtained within a few iterations.

Model Order Reduction of Finite Element Models of Electromagnetic Structures with Embedded Frequency-Dependent Multiports

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In an effort to tackle the complexity of electromagnetic computer-aided modeling at the subsystem and system level, domain decomposition techniques are often used. In such an approach portions of the system are identified, electromagnetic models of which can be reliably cast in terms of a multi-port transfer matrix function representation. Once these electromagnetic transfer functions have been computed, they serve as multi-port, frequency-dependent compact macromodels of the original physical structure. Subsequently, and for the purposes of system level modeling, these macromodels need to be re-introduced in the electromagnetic model of the entire system. Of interest here is the case where the system-level model is a finite element model. As a representative example of such a structure we mention the printed-circuit-board (PCB) portion of the power and signal distribution network of an integrated electronic system, which consists of multiple power and ground planes and multiple signal layers sandwiched between the planes. Assuming that multi-conductor transmission line (MTL) models can be used for the coupled interconnects in the signal distribution network (SDN), the complexity of the finite element modeling of such a system is simplified significantly. More specifically, the subsequent analysis can be carried out through the use of a finite element model for the power and ground planes of the power distribution network, with multi-port matrix transfer functions for the MTLs attached at the appropriate location where the MTL ports are defined.

The emphasis of this paper is on the development of a methodology for the direct generation of reduced-order macromodels of the hybrid models that are obtained from the aforementioned integration of frequency-dependent, multi-port macromodels in a finite element model of the remaining structure. The macromodels are assumed to be cast in terms of rational functions of the complex frequency $s = j\omega$. The proposed methodology utilizes Krylov subspace-based methods for the direct generation of compact (low-order) macromodels of the overall structure. Like in the application of Krylov subspace methods for the direct model order reduction of finite element models of packaging structures including lumped elements (H. Wu and A.C. Cangellaris, *IEEE Trans. Microwave Theory Tech.*, vol. 52, p. 2305, 2004), the resulting methodology allows for the construction of the overall multiport macromodel over a broad frequency interval, at essentially the cost of a single frequency point solution of the finite element system.

The proposed methodology will be presented and its validity and efficiency will be demonstrated through its application to a few generic structures encountered in the design of integrated microwave planar circuits and high-speed digital signal and power distribution networks.

Session 11 — Time domain FEM. Part 2

Organizers

Hesthaven, J. S., Rylander, T.

Presentations

1. Simulating the Behavior of Metamaterials and their Applications: Time Domain Advantages and Disadvantages
   Ziolkowski, R. W.

2. A Novel Time Domain Approach for Modeling the Coupling Problem between Two Large Arrays Mounted on a Complex Platform and Separated by a Large Distance
   Mittra, R.; Abd-El-Raouf, H. E.; Huang, N.-T.

3. Stable interfaces between the time-domain FEM and the FDTD scheme
   Degerfeldt, D.; Rylander, T.

4. Yee's Scheme for the Solution of Maxwell's Equation on Unstructured Meshes
   Hassan, O., Sazonov, I.; Morgan, K.; Weatherill, N. P.

SIMULATING THE BEHAVIOR OF METAMATERIALS AND THEIR APPLICATIONS: TIME DOMAIN ADVANTAGES AND DISADVANTAGES

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Metamaterials are artificial materials whose electromagnetic responses can be engineered for a variety of applications. They are often created by incorporating in a periodic manner various types of artificially fabricated and extrinsic inhomogeneities in a finite-size background substrate. The dimensions of these inhomogeneities are usually much smaller than the operating wavelength, e.g., $\leq\lambda/10$, to obtain an effective medium.

We are investigating the possibility of matching electrically small antennas to free space by surrounding them with single negative (SNG) and double negative (DNG) cylindrical and spherical metamaterial shells. These antenna systems are traditionally very poor radiators. We have demonstrated that these metamaterial-based electrically small antenna systems can be made to be highly resonant, substantially increasing the power radiated by them. We have also demonstrated that the reciprocal scattering structures have resonantly large total scattering cross sections. Multilayered shells have been considered to enhance the frequency versatility of these antenna systems; active layers have been considered to engineer their dispersion properties to enhance their bandwidths. We have considered artificial magnetic conductors integrated with small low profile antennas and have demonstrated gigabit sequence transmissions with these systems. We have demonstrated very sub-wavelength laser cavities in the optical domain.

The ability to model effectively these electrically small systems in the presence of rather complex media poses a number of computational electromagnetics (CEM) challenges. We have primarily modeled many of the metamaterials, represented by homogeneous, non-dispersive media, and these antenna applications in the frequency domain with finite element (FEM) techniques. These frequency domain simulations have allowed us to study the characteristics of the specific resonances that are driven by the sources. On the other hand, we have modeled a variety of dispersive metamterials and their source and scattering applications with the finite difference time domain (FDTD) approach and the finite integration technique (FIT). These time domain approaches have allowed us to investigate additional issues including causality; the delay-times associated with the formation of the negative parameters and the corresponding metamaterial effects on communication signals; and the frequency bandwidths of the radiating systems in the presence of realistic, dispersive passive and active metamaterials. Specific CEM issues associated with modeling these metamaterial applications and the contrasts between the frequency domain and time domain simulations will be discussed in the presentation.

Finally, it is a challenge to design and fabricate the SNG and DNG metamaterials with engineered values of the relative permittivity and permeability in the frequency regions of interest for all of the noted applications. Several RF circuit-based and lumped element implementations to achieve the ε and μ values required for the MTM-based efficient electrically small antenna systems, i.e., unit cell sizes $\leq\lambda/100$, have been considered. These results and the CEM issues in obtaining them will also be given in the presentation.

A Novel Time Domain Approach for Modeling the Coupling problem between Two Large Arrays Mounted on a Complex Platform and Separated by a Large Distance

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Historically, various asymptotic methods, *e.g.*, the PO, GTD and PTD have been employed for the solution of electrically large electromagnetic problems. It is well known, however, that asymptotic techniques are not well suited for handling problems that involve inhomogeneous media, complex geometries and multiscale features, *e.g.*, phased arrays, and it is necessary to employ numerically rigorous techniques, such as the Finite Element (FE) method. Solution of large problems using a numerically rigorous approach is always challenging because of the heavy burden they impose on the CPU time and memory. In this paper we introduce a novel time domain method for solving large electromagnetic problems, *e.g.*, coupling between two large arrays separated by a long distance and located on a platform with complex material coatings--a problem that may be difficult if not impossible to handle on available computing platforms via a direct application of the available CEM techniques because of its large size and complex nature.

The proposed technique takes a cue from the principles of Time Domain Reflectometer (TDR) operation, in which we track the propagation of a signal in a device as a function of time. It is based on dividing the original problem into smaller sub-regions of manageable size along the "effective" direction of propagation, and evaluating the solution that are localized in each of these sub-regions by capturing essentially all the near-field interactions. The choice of the subdomain size is based on the available computing resources, and most of the practical problems of interest require the use of parallel platforms because the size of the subregions can be on the order of 10 λ, or even larger. The excitation of the sub-regions can either be direct sources, as for instance the feeds in the aggressor array, or be derived from the fields propagating into the domain from adjacent regions through the interfaces.

The procedure is initiated by solving the problem in the sub-region in which the aggressor array is located, typically over a region that is sufficiently large to capture the near-field interactions. The time signatures at the points located on the interface between the first region and its immediate neighbor are stored in a data file after down-sampling, and are subsequently used as effective sources for the region immediately to the right of the first one (assuming that the propagation of the coupling signals between the two arrays is from left to right). The process is continued until we reach the victim array and have solved for the induced voltages and currents in the elements of that array to estimate the EMI effects.

The paper will show that the results obtained by using the proposed approach compare well with the direct solution for several test problems involving coupling between two phased arrays. Of course, the advantage of using the proposed approach is that it can handle very large problems, involving 10^9 or even larger number of unknowns--well beyond the scope of the direct methods-- and with virtually no limit on the size of the coupling problems of the type described above that it can tackle.

Application of the method to EM scattering and EMI problems will also be mentioned.

Stable interfaces between the time-domain FEM and the FDTD scheme

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The finite element method (FEM) on unstructured grids of tetrahedrons is a powerful tool for the analysis of field problems with complex geometry in microwave engineering. Its time-domain formulation is favorable for broad frequency band analysis and non-linear problems. However, the computational cost for the time-domain FEM formulated on unstructured grids is relatively high as compared to closely related techniques such as the finite-difference time-domain (FDTD) scheme. In fact, the FDTD scheme can be interpreted as an explicit FEM on brick elements with lumping for both the "$\nabla \times \mu^{-1} \nabla \times$"– and "$\epsilon\, \partial_t^2$"–operators. This is a useful perspective for the construction of hybrids that combine explicit time-stepping on bricks with implicit time-stepping on unstructured grids.

One possible solution is to use pyramidal elements to connect the brick FDTD cells with unstructured tetrahedrons, which gives a curl-conforming representation when the electric field is expanded in edge elements. Galerkin's method applied to the self-adjoint Maxwell's equations yields real and non-negative eigenvalues. In turn, this makes it possible to formulate a time-stepping algorithm that is stable for time steps up to the stability limit of the FDTD without added dissipation. The hybrid scheme is free from spurious solutions. The bistatic radar cross section for a metal sphere gives 1 dB accuracy with only 9 cells per wavelength and it shows second order convergence with respect to the cell size.

Recently, we have developed a new hybrid that connects the tetrahedrons directly to the FDTD cells. This simplifies the construction of hybrid grids significantly in the sense that the pyramids can be avoided. The new hybrid has all the important features: efficiency, accuracy, stability, no spurious solutions and expected order of convergence. The talk will address this type of hybrids in a comparative manner and illustrates the performance differences in terms of accuracy and efficiency for some canonical problems.

Yee's Scheme for the Solution of Maxwell's Equation on Unstructured Meshes

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Abstract

Yee's scheme [1] is a very fast and accurate method for the computation of the scattering of electromagnetic waves. This algorithm is known for its low operation count, as it requires few arithmetic operations to advance the solution in time, particularly when it is compared, for example, to the Time Domain Finite Element Method (TDFE). Initially proposed for structured grids, Yee's scheme can be generalized for unstructured meshes. This will enable its application to industrially complex geometries [2].

Yee's scheme is a staggered algorithm both in space and time. It needs two orthogonal meshes for the electric and magnetic fields. A dual Delaunay-Voronoi diagram forms such two orthogonal meshes. Since the time step limitation of Yee's algorithm is proportional to the shortest edge in the both dual meshes, it is critical that the Voronoi diagram is as well behaved as the Delaunay triangulation.

Using present meshing methods it is possible to build a smoothly varying Delaunay triangulation, with a high quality index. However, the resultant Voronoi diagram is highly irregular, with some very short Voronoi edges. This will increase the total number of required time steps per cycle and hence reduce the practicality of the method.

The approach proposed here will allow the generation of unstructured meshes in which the lengths of the Delaunay and Voromoi edges are bounded from above and below. First, a high quality near-boundary triangulation is prepared. A structured mesh of equilateral triangles is used to fill the remainder of the domain. The near-boundary triangulation is 'stitched' to the ideal mesh so that the 'seam' is also of high quality.

Computations using meshes generated by this method for different geometries will be presented. The radar cross section (RCS) will be compared to exact and other computed solutions. The efficiency and accuracy of the method will also be analysed and compared to the efficiency of a finite elements time domain method. Geometries which contain singularities will also be used to demonstrate the advantages of the proposed scheme.

The extension of the proposed approach into three dimensions will also be addressed and initial results which demonstrate the efficiency of the scheme will be shown.

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Session 12 — Hybrid Methods

Organizers

Lee, R.; Volakis, J. L.

Presentations

1. A Non Standard Fast Multipole Finite Element Method for Scattering and Radiation Problems
   Barrio-Garrido, R. M.; García-Castillo, L. E.; Gómez-Revuelto, I.; Salazar-Palma, M.

2. On the Formulation of the 3D Hybrid Finite Element - Boundary Integral Technique: … Still the Old One: Formulation, Solution, Results
   Eibert, T. F.; Tzoulis, A.

3. Hybrid MoM-FDTD for Human RF Exposure Analysis
   Lacroux, F.; Wong, M.-F.; Wiart, J.

4. Modeling Large Phased Array Antennas Using the Finite Difference Time Domain Method and the Characteristic Basis Function Approach
   Mittra, R.; Farahat, N.; Huang, N.-T.

A Non Standard Fast Multipole Finite Element Method for Scattering and Radiation Problems

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The Finite Element Method (FEM) is a powerful and flexible numerical technique with successful application in different areas of physics, including electromagnetics [1]. However, FEM formulation does not incorporate the radiation condition. Thus, when dealing with open region problems, as in the case of scattering and radiation of electromagnetic waves, an artificial boundary must be used to truncate the problem domain (mesh) in order to keep the number of unknowns finite. Several approaches can be used to make the artificial boundary "transparent" to electromagnetic waves. The authors have developed a FEM methodology for open region problems [2], [1], [3], [4], that imposes a numerically exact radiation boundary condition through the use of the Integral Equation (IE). However, in contrast with standard FEM-BI approaches, it is done in such a way that the original sparse and banded structure of the FEM matrices is retained, allowing the use of efficient FEM solvers. This is achieved at the expense of performing a small number of iterations in which only the right hand side of the algebraic system of equations changes. The right hand side is associated with the residual of a local boundary condition (specifically, of Cauchy type) and is updated using the IE representation of the exterior field (i.e., in terms of the Green's function of the exterior problem). Actually, the IE is considered on an auxiliary boundary which is interior to the truncation boundary. The use of efficient FEM solvers makes the bottleneck of the method to be transferred to the computation of the field using the IE and the Green's function of the exterior problem. Several methods have appeared in the literature to accelerate those computations. In this paper the Fast Multipole Method (FMM) [5] is applied to the non standard FEM for open region problems described above. The implementation of FMM presents some peculiarities due to the non standard character of the FEM methodology in which it is integrated. Several numerical results of the application to scattering and radiation problems of a two dimensional implementation of the FMM-FEM method proposed will be shown.

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On the Formulation of the 3D Hybrid Finite Element - Boundary Integral Technique: … Still the Old One: Formulation, Solution, Results

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The hybrid finite element – boundary integral (FEBI) technique has proven to be a very powerful method for the numerical computation of electromagnetic radiation and scattering problems. Its greatest advantages are the relatively small disrectization effort compared to purely local techniques, the integration of BI (MoM: method of moments) models of metallic objects, and the efficient treatment of mixed metallic/dielectric objects as well as of various objects not directly connected to each other. Over the past years, there was a lot of discussion on the optimum formulation used for the realization of 3D FEBI techniques. All of these formulations generate the necessary equations from electric and/or magnetic FE functionals and the electric and/or magnetic field integral equations, where different formulations combine the equations in different ways and work with different basis and testing functions. Our formulation generates one set of linear equations from a standard FE functional in terms of the electric field intensity and a second set of linear equations from the combined field integral equation (CFIE), where lowest order edge element basis functions are used in the FE and Rao-Wilton-Glisson basis and testing functions are employed in the BI. Such a procedure results in a hybrid equation system with "well-tested" matrix blocks along the main diagonal, where the off-diagonal blocks realizing the coupling between FE and BI subsystems could be called "not well-tested", if they were considered as stand-alone matrices. Our experience is, that the solution of these equations provide accurate results for radiation and scattering problems. Iterative solution of the equation systems for practical problems is certainly not easy, but we obtain good convergence with a recursive iterative solution strategy based on a flexible Generalized Minimal Residual Solver (GMRES). Utilizing the Multilevel Fast Multipole Method (MLFMM) for the evaluation of the BI, large-scale problems can be solved in an efficient manner, where also the near-fields and far-fields generated by the BI currents are computed by MLFMM. A variety of computational results and comparisons to measurements are shown.

Hybrid MoM-FDTD for human RF exposure analysis

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With the advent of mobile communications, public concerns have raised on the possible hazards of radiowaves. International bodies such as International Commission on Non-Ionizing Radio Protection (ICNIRP) or IEEE have defined safety limits to protect people. Fundamental limits are given in terms of Specific Absorption Rate (SAR). Derived limits are also provided in terms of electric field strength to help assessment. In the case of base station antennas used in cellular networks, simulations can be used to compare the compliance boundaries (i.e. surfaces where the limit is reached for a given emitted power) induced by the fundamental and derived limits. The derived limits give more conservative compliance boundaries. The margin between the two kinds of limits is particularly of interest. Many calculation techniques can be used; we propose here a hybrid one which couples MoM and FDTD. FDTD is widely used in bio-electromagnetics as it is able to realistic heterogeneous human models coming from MRI. Hybridizing FDTD with MoM via a Huygens box is interesting when the distance between the source and the body is increased in parametric studies. No meshing of the filling space is required unlike a standard FDTD calculation and thus computation resources are saved. Furthermore, MoM can be more appropriate for the modeling of the antenna. When increasing the distance, the system becomes more and more decoupled. One-way coupling is analyzed by comparing to full coupled results. The technique is applied to the case of UMTS base station antennas. The relation between the electric field strength and whole-body and localized SAR are discussed.

Modeling Large Phased Array Antennas Using the Finite Difference Time Domain Method and the Characteristic Basis Function Approach

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Numerical modeling of large but finite phased array antennas is a challenging problem because it places a heavy burden on the computer resources, especially when the array element is complex, and the antenna operates in a close proximity of an FSS radome whose period is not commensurate with that of the array. The array element is typically a microstrip patch, a Vivaldi or a waveguide, and the antenna may be covered by an FSS radome, whose elements may be patches, slots, cross-dipoles, *etc.*, typically different from those of the array. The array can also be mounted on a curved surface and this presents an additional challenge because the unit cell approach, commonly used for planar, doubly-periodic infinite arrays can no longer be used to analyze such conformal arrays. Accurate prediction of the performance of such complex antenna systems is a very challenging problem indeed.

In this paper we describe an approach for solving large phased array problems, using the domain decomposition approach, combined with the Characteristic Basis Function Method (CBFM). The method is especially tailored for analyzing arrays that may be covered with Frequency Selective Surfaces (FSSs), especially when the array and FSS periods are not commensurate.

The first step in this approach is to define a sub-domain using a stencil, within which the local interactions are essentially confined. The size of the stencil may have to be several wavelengths in order to capture these near-neighbor interactions. The solution to the original problem is then synthesized by using the solutions to these subdomain problems that are manageable on the available computing platforms.

Several illustrative examples will be provided in the presentation, which will include validating results for a number of test cases. The validation will be accomplished by comparing the results derived by using the proposed technique with those obtained via a direct simulation of the entire array on a PC cluster. Of course, the direct problem places a heavy demand on the CPU memory and time--especially as the problem size becomes large--and may in fact be intractable on the available computer resources. We will show that, in contrast to the direct method, the increases in the simulation time and the burden on the computer memory are incrementally small in the present approach, as the problem size is increased from moderate to large, even when the size of the array becomes very large.

Session 13 — Advances in FEM Formulations and Solvers

Organizers

Botha, M. M.; Jin, J.-M.

Presentations

1. Homogenization of Large Screens in Finite Element Analysis
   Bardi, I.; Cendes, Z. J.

2. SPICE-Compatible Finite Element Models of Maxwell's Equations
   Cangellaris, A. C.

3. An Efficient Solver for High-Order Vector Finite Elements
   Ingelström, P.; Hill, V.; Dyczij-Edlinger, R.

Homogenization of Large Screens in Finite Element Analysis

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Electronic equipment often contains screening grids where design engineers need to determine the shielding effectiveness of perforated grids in the walls of electronic cabinets. Typical examples are establishing EMI/EMC radiation leakage levels arising from lattices of ventilation holes in computers and other electronic equipment. Electromagnetic interference can emanate through these lattices which can perturb the operation of the device or the device can perturb the operation of other equipment in the environment. The simulation complexity of electronic equipment can be reduced considerably if the ventilation grids/screens are replaced by electromagnetically equivalent continuous sheets. The procedure of replacing repetitive small structure geometric patterns by equivalent continuous materials is called homogenization in the literature. In this paper, the geometric patterns are replaced not by an equivalent material property but by an equivalent surface impedance. The homogenized characteristic surface impedance is determined by modeling the behavior of a unit cell of a periodic grid of infinite extent. While this approximation neglects the edge effects of the grid, we will show by means of examples that this approximation provides practical accuracy when the repetition number in the grid is large.

Consider the simple metal cabinet shown in Figure 5. The dimensions of this cabinet are 300 x 300 x 150 mms. One side of the cabinet is a ventilation grid consisting of a lattice of 25 x 25 holes. The total grid area is 30 x 30 mms. A Hertzian dipole is placed in the middle of the cabinet and excited at 3 GHz. The far field of the structure was solved twice: once by modeling the grid in complete detail and a second time by replacing the grid with the homogenized impedance boundary condition. At 3 GHz, the impedance is j86 Ohm. The far field of both the control and the homogenized project is shown in Figure 6. The results show a good agreement.

Consider now a real-life computer cabinet with several ventilation grids, a conducting gasket and other details. In this case, the excitation is from the real-life printed circuit board (PCB) shown in Figure 9. The current distribution in the PCB is first computed by using a full wave finite element procedure specialized to solve planar structures. The currents give rise to a near-field distribution which is passed on to the 3D tool that handles the computer housing. Subsequently the fields inside the cabinet and the radiated fields through the ventilation grids are computed. The resulting radiated field is shown in Figure 10. In that figure, the PCB is not visible, as its presence is handled through the dynamic link with the planar method.

The 3D field simulation of the computer housing required an initial mesh of 105,000 tetrahedra when all the ventilation holes were present explicitly. This was reduced to 21,000 tetrahedra when most of the ventilation holes were replaced by impedance boundaries. Hence, the number of unknowns was reduced by a factor of five. The radiated fields are very similar in both approaches.

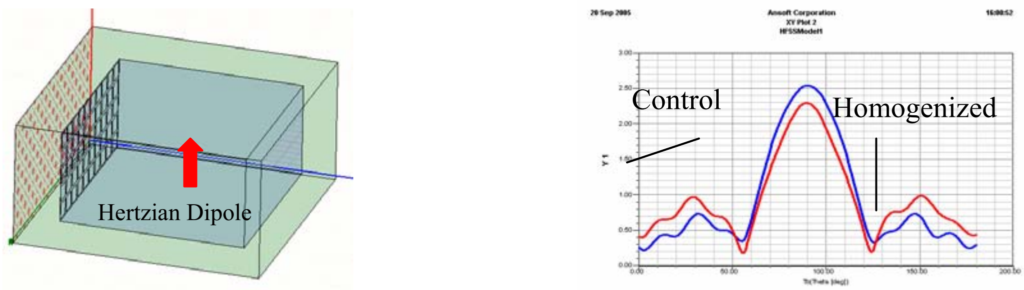

Figure 6. Hertzian dipole in a metal cabinet with ventilation grid and far field.

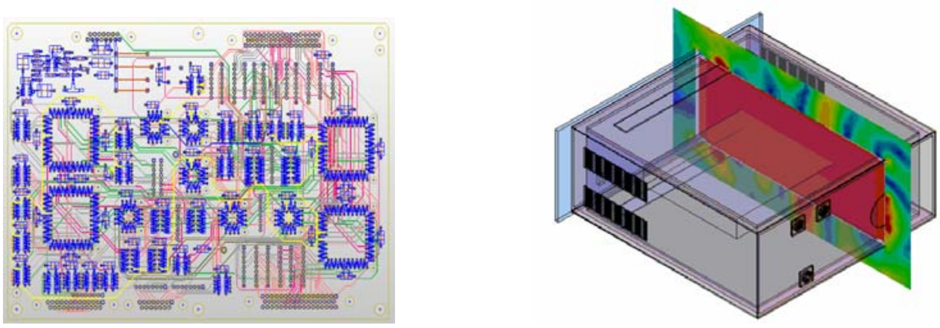

Figure 9. Radiation through ventilation screens from the PCB on the left placed in the computer cabinet.

SPICE-Compatible Finite Element Models of Maxwell's Equations

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In order to provide for concurrent electromagnetic/circuit simulation, most hybrid, time-domain electromagnetic/circuit simulators utilize the discrete electromagnetic model, resulting from the spatial discretization of Maxwell's equations on a finite difference or finite element grid, for the incorporation of lumped circuit element-based models of electronic devices and/or electrically-small passive electrical components. In such a modeling approach, the electromagnetic field solver becomes the underlying simulation framework, with the voltage/current state variables in the non-linear system of time-domain equations governing the behavior of the lumped circuit components being interfaced with the discrete electric and magnetic fields in the grid through appropriately-defined line integrals.

While such an approach is most appropriate for those cases for which the lumped (circuit) components constitute an electrically-small portion of the overall electromagnetic structure of interest, the reverse approach, namely, the incorporation of a semi-discrete electromagnetic model for a portion of the structure under investigation inside a circuit-based model of the structure is also useful, particularly in relation to the modeling of multi-physics phenomena involving multiple spatial scales of large disparity.

To provide for the incorporation of the semi-discrete (state-space) model, resulting from the spatial discretization of Maxwell's equations in general-purpose, non-linear circuit simulators such as SPICE, this paper proposes a methodology that leads to the representation of the semi-discrete systems in terms of stamps for the differential equations governing the temporal evolution of the discrete electric and magnetic field components. To provide for modeling versatility, the proposed methodology utilizes the finite element approximation of Maxwell's curl equations on a finite element mesh of tetrahedrons. Through the use of edge elements for the expansion of the electric field intensity and normally-continuous "flux" elements for the finite element approximation of the magnetic flux density, a semi-discrete model results which can be expressed in terms of SPICE-compatible stamps with readily recognizable interpretations in terms of lumped circuit elements, coupled inductors and dependent sources.

The presentation will consist of an explanation of the methodology used for the development of the stamps and the validation of the resulting SPICE-compatible description of the semi-discrete electromagnetic system. In particular, it will be shown that there is a very close relationship between Maxwell's curl equations and the mesh analysis formalism of Kirchhoff's equations which can be readily utilized for identifying the appropriate state variables to be used for the SPICE-compatible description of the semi-discrete (state-space) form of Maxwell's equations.

An Efficient Solver for High-Order Vector Finite Elements

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We discuss some issues regarding efficient solvers for high-order H(curl)-conforming elements. This includes both the type of solver, the type of basis functions and implementational details, including how to store – or not to store – the system matrix.

The solver we use is a Krylov subspace solver with a multiplicative Schwarz preconditioner based on hierarchical basis functions. This type of solvers has become very popular for second order schemes in recent years. The system matrix is divided into blocks according to the polynomial degree of the basis functions. A complete factorization is used for the first order block whereas the other diagonal blocks are inverted approximately using forward/backward Gauss-Seidel relaxation. As the block size increase with the order, operations on the higher order blocks are more expensive. To reduce this cost, we present a W-cycle scheme that operates more frequently on the low-order blocks. Compared to the more standard V-cycle, the W-cycle reduces both the number of iterations and simulation time significantly.

There are several bases for tangentially continuous elements in the literature today. We use a recently developed basis [1] that has a few advantages compared to other available bases: Firstly, it leads to very sparse stiffness matrices (block tri-diagonal). Secondly, it gives good convergence with the solver mentioned above. However, many other bases that can be found in the literature gives very similar convergence for second and third order schemes.

A disadvantage of higher-order schemes is that the system matrices are denser than for low order schemes. Hence the amount of memory required to store the matrix in standard compressed sparse row (CSR) format easily grows very large. Another disadvantage of the CSR format is that the performance of matrix-vector multiplications is quite poor since it is limited by memory bandwidth rather than floating point operation capacity. However, to efficiently perform Gauss-Seidel relaxation, The CSR format or similar is advantageous. Hence, we use the CSR format for the diagonal blocks but present two alternative ways for storing and performing matrix-vector multiplication on the off-diagonal blocks. To maximize speed, we perform a global matrix-vector multiplications using the element matrices, This is not efficient for first order schemes, but for fourth order schemes we easily get a speed-up factor of two. If instead, the memory requirement should be minimized, we note that the shape of a tetrahedron can be described using six parameters only. Hence we can pre-compute twelve element-independent master matrices (six for the mass matrix and six for the stiffness matrix), see [2]. Even though the number of operations in increased by an order of magnitude, most of the data will remain in the cache memory all of the time. Hence we can make much better use of the computer's floating point operation capacity. Under favorable conditions we can actually get similar performance as for a matrix in the CSR format.

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Session 14 — Optimization Techniques, Parameter Space Sweep. Part 1

Organizers

Pelosi, G.; Zapata, J.

Presentations

4. Order Reduction of Finite Element Models Parameterized by Multivariate Polynomials
   Farle, O.; Hill, V.; Ingelström, P.; Dyczij-Edlinger, R.

5. A Reduced Order Model for the Rapid Computation of Monostatic Scattering Width Distributions
   Ledger, P. D.; Morgan, K.

6. Genetic Optimization of Unconventional Inductive Frequency Selective Surfaces by Using a Hybrid Mode Matching - FEM technique
   Pellegrini, A.; Monorchio, A.; Manara, G.

Order Reduction of Finite Element Models Parameterized by Multivariate Polynomials

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In computational electromagnetics, methods of model order reduction (MOR) are most commonly used for the calculation of frequency sweeps. In recent years, MOR has been extended to the multi-parameter case [1] [2], and, nowadays, model parameters may also include incident angles [3], wire spacings [4], or material properties.

We propose a new MOR method for systems of linear equations parameterized by polynomials in several variables. For N scalar parameters $s_1, s_2, \ldots, s_N$ and polynomial degree M, such systems take the form

$$\left(\sum_{|\alpha| \leq M} \mathbf{s}^{\alpha} \mathbf{A}_{\alpha} \right) \mathbf{x} = \sum_{|\alpha| \leq M} \mathbf{s}^{\alpha} \mathbf{b}_{\alpha}, \tag{1}$$

where $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_N)$ is a multi-index, and $\mathbf{s}^{\alpha}$ is defined as $\mathbf{s}^{\alpha} = s_1^{\alpha_1} s_2^{\alpha_2} \ldots s_N^{\alpha_N}$.

Our main contribution is to characterize the solution space of (1) by generalized Krylov subspaces. Based on this framework, we construct parameter-independent projection bases, which achieve moment-matching between original and reduced models and accommodate both parameter-dependent system matrices and right-hand sides. With the new approach, it is not necessary to transform (1) to an equivalent multi-linearly parameterized system of larger size.

Most multi-parameter MOR methods compute moments directly, which may lead to serious numerical cancellation. To overcome this deficiency, we present a recursion that operates on an orthogonal basis at any stage of the iteration. The proposed algorithm is capable of generating accurate reduced models of very high order. As a single-point method, it requires one matrix factorization only.

In the presentation, we shall develop the underlying theory in detail and present numerical examples to demonstrate the reliability of our approach.

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A Reduced Order Model for the Rapid Computation of Monostatic Scattering Width Distributions

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The foundation of any numerical approximation is a reliable and accurate discretization procedure. With this objective, we have recently devoted considerable effort into the efficient implementation of an $\boldsymbol{H}(\text{curl})$ conforming *hp* finite element algorithm for the solution of scattering problems governed by Maxwell's equations [2, 4, 3, 1].

By using the *hp* finite element approach, outputs of practically interesting quantities, such as the scattering width distribution, may be accurately computed. However, the interest of the designer lies in investigating the outputs over the complete parameter space and the computational cost of performing an *hp* finite element computation for each parameter value may be prohibitively expensive for practical purposes. Maday, Patera and co–workers [7, 6] have developed the adjoint enhanced reduced order modeling technique, which enables the rapid, efficient and accurate computation of engineering outputs for new values of the problem parameters. Ledger et al [5] demonstrated how this technique could be applied to the rapid prediction of the bistatic scattering width distribution for new incident wave directions, in two–dimensional Maxwell scattering simulations, employing an implementation with the added value of computable certainty bounds, which guaranteed the level of accuracy of the prediction.

In this talk, we will describe the application of the adjoint enhanced reduced order modeling technique to the rapid computation of monostatic scattering width distributions. The computation of monostatic distribution is, in general, more expensive than the computation of bistatic distribution, as the monostatic profile requires a finite element simulation to be performed for each viewing angle of the coordinate system. The use of direct solvers with multiple right hand side vectors can reduce the computational cost although, with three–dimensional simulations in mind, the cost of computing such distributions remains expensive. In this initial study, we shall discuss how two–dimensional monostatic computations can benefit from the incorporation of the adjoint enhanced reduced order model and remark upon how the technique can be extended to handle the full three–dimensional case.

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Genetic Optimization of Unconventional Inductive Frequency Selective Surfaces by Using a Hybrid Mode Matching - FEM technique

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An optimization procedure for inductive Frequency Selective Surfaces (FSSs) is presented, which is based on the use of the Genetic Algorithm (GA). The GA synthesis procedure calls for an electromagnetic solver based on the combination of the Mode Matching (MM) method and the Finite Element Method (FEM). It allows us to efficiently analyze inductive FSSs with arbitrarily shaped apertures in a thick metallic screen. In particular, in the present method we exploit an advantage of the evolutionary algorithm, which does not require the *a priori* definition of a first guess structure. Moreover, due to the numerical efficiency and accuracy of the Mode Matching and to the flexibility of the FEM analysis, the GA is allowed to investigate and explore unconventional solutions that can provide better performance than the conventional ones. A key step in the design procedure consists of a suitable choice of the parameters that control the optimization procedure. For the problem at hand, we select these to be the dimensions of the unit cell along the principal periodicity directions, the thickness of the metal plate and the shape and size of the apertures. To illustrate the procedure, let us consider a plane wave impinging on the FSS along an arbitrary angle. As a first step, the electromagnetic field in the free-space region is expanded in a complete set of Floquet modal basis functions, with unknown complex coefficients. Next, the transverse electromagnetic fields inside each arbitrarily shaped aperture, considered as a section of a metallic waveguide, are represented by a complete set of waveguide modes obtained via the FEM approach. The occurrence of spurious solutions is avoided by the use of Whitney's Edge elements to interpolate the transverse field components. It is important to note that the mesh generation needed at this point must be fast and accurate. For this reason, in order to get a good control over this critical point, a 2D-Quality mesh generator has been developed and embedded into the electromagnetic analysis software. This tool is based on the Boyer-Watson approach to Delaunay triangulation. In fact, Delaunay properties guarantee a mesh with pseudo-equilateral triangular finite elements in order to achieve a good interpolation of the fields. The adaptive mesh enable us to easily handle particularly irregular geometries, representing a favourable compromise between numerical accuracy and computing time. Through the computation of the unknown modal coefficients, the Generalized Scattering Matrix (GSM) of the free-space/waveguide interface is determined. Finally, the GSM of the entire screen is obtained by considering the cascade connection of the lit face, the aperture section, and the shadow face, then combining the corresponding matrices in accordance with the standard formulas for microwave circuits. The electromagnetic analysis described above is used to derive the transmission and reflection coefficients; they are employed to evaluate the fitness function of every single structure needed at each generation of the GA. It is important to highlight that this efficient and flexible methodology enable us to explore a much more extensive range of solutions provided by the unconventional shapes. Some numerical results will be presented at the conference to show the effectiveness of the optimization procedure.

Session 15 — Multigrid- and Domain Decomposition Methods. Part 2

Organizers

Cangellaris, A. C.; Lee, J.-F.

Presentations

1. Adaptive Domain Decomposition Techniques in Electromagnetic Field Computation
   Hoppe, R. H. W.

2. A Nonuniform hp-Finite Element Method for the Computation of Electromagnetic Cavity Modes
   Nickel, P.; Hill, V.; Ingelström, P.; Dyczij-Edlinger, R.

3. Studies of Numerical Error Introduced by the Cement Variables in the Domain Decomposition Method
   Rawat, V.; Lee, J.-F.

Adaptive Domain Decomposition Techniques in Electromagnetic Field Computation

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Abstract

We consider adaptive multilevel and domain decomposition methods in the numerical solution of interior domain problems for the time-harmonic Maxwell equations and the eddy current equations in three space dimensions. Based on a geometrically conforming decomposition of the computational domain into non-overlapping subdomains and adaptively generated hierarchies of locally quasi-uniform and shape regular simplicial triangulations, the subdomain problems are discretized by the lowest order edge elements of Nédélec's first family. Special consideration is devoted to mortar edge element techniques in case of non-matching grids on the interfaces between subdomains. The numerical solution features a multilevel preconditioned iterative solver, whereas mesh adaptivity is realized by means of a residual-type a posteriori error estimator that relies on a Helmholtz decomposition allowing a separate estimation of the irrotational and weakly solenoidal part of the error.

Applications include the numerical simulation of LWD (<u>L</u>ogging <u>W</u>hile <u>D</u>rilling) tools used in oil exploration for in situ measurements of geological formations and the structural optimization of electromotors in high power electronics.

A Nonuniform hp-Finite Element Method for the Computation of Electromagnetic Cavity Modes

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The finite element (FE) method offers two complementary ways to improve the quality of approximation: one may increase the polynomial order of the basis functions p or shrink the element size h. While h-refinement is always limited to algebraic rates of convergence, p-enrichment may yield exponential convergence, provided that the fields are sufficiently smooth. However, when singularities are present, p-schemes perform poorly, whereas nonuniform h-methods manage to keep the rate of convergence unchanged. Since real-world cavities often comprise both regions of smooth and singular fields, nonuniform hp-methods are very attractive.

The proposed FE method is based on p-hierarchical basis functions [1] together with nested mesh refinement and hanging variables [2] for stitching unequal h-levels. Specifically, we construct a hierarchy of nested FE discretizations by applying geometric multigrid (GMG) and local sub-gridding techniques. The p-hierarchical FE basis makes it straightforward to vary the polynomial degree as appropriate.

For solving the algebraic eigenvalue problem, we employ a variant of the Jacobi-Davidson (JD) algorithm [3]. JD methods require low quality solutions to the correction equation only, exhibit convergence rates that are quadratic at least, and are easily generalized to polynomial eigenvalue problems.

Throughout the solution process, we exploit the nested structure of our FE spaces in two different ways. First, we apply fast GMG and p-multilevel techniques to solve the JD correction equation. Second, we adjust not only the final residual of the iterative solver in accordance with the JD iteration count [3], but also resolution of the FE space.

In our talk, we shall give the details of the proposed method, including our restart and deflation strategies. To demonstrate the efficiency of the suggested approach, we shall present numerical examples that involve complicated geometries and large numbers of unknowns.

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Studies of Numerical Error Introduced by the Cement Variables in the Domain Decomposition Method

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The Domain Decomposition Method (DDM) has been successfully applied to the solution of problems that are otherwise intractable when the conventional Finite Element Method (FEM) is used. Many electrically large problems are composed of multiple repeating sub-structures and lend themselves naturally to a decomposition exploiting this redundancy. To model large finite periodic structures, the DDM requires the identification of repeating sub-structures, or building blocks, and a decomposition of the region of interest into sub-domains based on these blocks. In FEM solution, the computation associated with a block is re-used, yielding a significant reduction in required resources.

The decomposition of the domain of interest and subsequent FEM meshing within each sub-domain leads to two closely associated problems; a global, or coarse problem exists over the large mesh of sub-domains and a local, or fine problem arises via the traditional FEM treatment within each sub-domain. The convergence rate of the stationary or Krylov subspace iterative solver used in the global problem solution depends critically on the enforcement of boundary conditions at interfaces between sub-domains. This, in turn, is related to the quality of the solution on interfaces, whose error is associated with the FEM meshes of adjacent sub-domains. The characterization of this relationship is a key issue in achieving efficient operation of the DDM and is the focus of this presentation.

The propagation of information between sub-domains in the DDM is accomplished via two mechanisms on the coarse mesh: cement variables are used to “glue” adjacent sub-domains together and novel higher-order transmission conditions that use these cement variables enforce the appropriate boundary conditions between sub-domains. The cement element technique for the DDM splits an interface between adjacent sub-domains into two distinct surfaces, one belonging to each sub-domain. Corresponding variables representing physical quantities are then introduced on each of these new surfaces, allowing non-conformity between the FEM meshes of adjacent sub-domains. This allows the independent meshing of each sub-domain but comes at the expense of losing the strict enforcement of continuity in physical quantities at sub-domain interfaces. Higher-order transmission conditions employ the cement variables of adjacent domains in enforcing the boundary conditions on interfaces. The effectiveness of these conditions depends on the accuracy to which cement variables originating in one domain can be computed over non-matching elements in a neighboring domain; error introduced due to non-matching interface meshes will pollute the enforcement of transmission conditions and compromise their accuracy.

Here, we study specifically the effect of refining the size of the fine FEM mesh on interfaces (h-type mesh refinement) as well as the use of higher order basis functions on interface elements (p-type refinement).

Session 16 — Optimization Techniques, Parameter Space Sweep. Part 2

Organizers

Pelosi, G.; Zapata, J.

Presentations

1. FEM Optimization of Coplanar-to-Coplanar waveguide transitions in Flip-Chip Interconnections
   Agastra, E.; Limiti, E.; Pelosi, G.; Selleri, S.

2. Optimization of Devices Based on Bodies of Revolution with Finite Elements
   Martínez-Fernández, J.; Monge, J.; Zapata, J.; Gil, J. M.

3. Parametric FEM Modal Analysis in microwaves and EMC
   Wong, M.-F.; Gati, A.; Orjubin, G.; Richalot, E.; Picon, O.; Wiart, J.; Hanna, V. F.

FEM Optimization of Coplanar-to-Coplanar waveguide transitions in Flip-Chip Interconnections

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Increasing interest in high frequency mobile sensor and wireless communications applications demands for low cost packaging solutions with excellent performances even at millimeter-wave frequencies.
Among the different possibilities for multiple monolithic millimeter wave integrated circuits (MMIC), providing different functionalities, to be mounted over a single substrate board there is the flip-chip scheme. In this approach multiple MMIC are ‘flipped’ and connected to the substrate, that is mounted upside down, with small soldered bumps providing electric contact [1].
A full comprehension of the effects of such mounting is of course vital to achieve an efficient design process. The flip-chip mounting can indeed produce several parasitic effects, the most relevant of which are: the detuning of the circuit on the chip due to its proximity to the motherboard [2] and the reflection at the bump interconnect itself, due to the abrupt transition between the guiding structure on the substrate and on the chip [1]. On the top of these two the package may introduce additional parasitics, primarily with regard to surface waves on substrate, and internal resonances.
This contribution is aimed at the study of the reflection coefficient of a flip-chip mounting exhibiting both a GaAs chip and board, with gold conductor backed Coplanar Waveguide (CPW) as guiding structures on both. The main difference between some of the other authors, for example [1], is the usage of a Finite Element Method (FEM) [3], allowing not only for an exact description of the basic flip-chip mounting, with correct evaluation of cylindrical/spherical bumps, but also allowing for the evaluation of “perturbed” geometries, with the chip not exactly centered in place (translation tolerances) and not aligned with (rotation tolerances). This latter analysis is indeed impossible with many other methods, like, for example, Finite Differences - Time Domain (FDTD). On the basis of these consideration an optimal design is sought for, that is a design wich is the least sensitive to these tolerance effects.

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Optimization of Devices Based on Bodies of Revolution with Finite Elements

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This paper deals with the optimization of radiating electromagnetic structures based on bodies of revolution. To perform the optimization an adaptative version of the Simmulated Annealing (SA) algorithm [1] has been used. For each analysis in the optimization loop, the scattering matrix of the structure is obtained by an efficient technique based on the Segmentation technique, the Finite-Element and a Matrix Lanczos Padè algorithm (SFELP), adapted to problems with axial symmetry. To obtain the radiation pattern, a propagation of spherical modes defined in a spherical port that encloses the domain is evaluated. Designs obtained from this technique are presented and range from a special type of monopole to some types of conical horns.

Devices based on bodies of revolution have the advantage of its symmetry that simplifies the manufacture process. Canonical forms as conical profiles not always satisfy specifications so corrugated horns, dielectric loaded conical horns, as well as enhanced monopoles have been studied in an attempt to correct a specific parameter of the antenna, or to try to reduce the size and weight of the structure.

We will present arbitrarily shaped profile horns loaded with dielectric designed to maximize its apperture eficency. Other objetives like a specific tapper of an empty horn used as a feed of a reflector have also been tested. An ultrawideband short monopole has been designed by performing an optimization on the profile of the monopole with several curved steps. All this designs are optimized with SA and SFELP [2] taking into account mechanical constrains.

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Parametric FEM Modal Analysis in microwaves and EMC

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As the Finite Element Method (FEM) applied to solve Maxwell equations has proven its ability in analyzing more and more complex microwave devices, the natural step forward is towards FEM oriented design and optimization. Amongst powerful techniques to achieve that goal are model order reduction techniques. Many approaches have been developed. The common idea between the different techniques is that the frequency response of a electromagnetic system can be efficiently described by a finite and small number of spectral components, i.e. eigenvalues and eigenvectors. These techniques are also known as fast frequency sweep ones as the response over a wide frequency band can be obtained at the cost corresponding to the solving of the problem at a few frequencies. A 3D FEM based modal analysis has been developed where the forced oscillation solution of a microwave N-port can be decomposed into the modes, i.e. free oscillation solutions, of the unexcited N-port. It leads to a reduced order model in terms of residues and poles that can be used as macro-model for both time and frequency simulations. This technique can be extended to tackle shape variation of the system leading to wideband models parameterized in geometry. The modes are parameterized in geometry using a Taylor expansion where the derivatives with respect to geometric parameters can be determined recursively. The calculations require a parameterized mesh that is used to get the derivatives with respect to geometric parameters of the FEM matrix. The proposed technique has been successfully applied to carry out parametric analysis of microwave structures, design of filters and eventually the analysis stirred mode reverberating chambers used for electromagnetic compatibility test and evaluation. In this last example, an essential feature of the technique is demonstrated. The variations of the resonant frequencies are free of numerical noise which may occur when direct computations are made on different meshes corresponding to sample values of the geometric parameters.

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